UPSC Prelims

Environment & Ecology

Class No. 12

India’s Initiatives

India’s Initiatives

India's National Action Plan on Climate Change (NAPCC)

- NAPCC was published in **2008** by the then-Prime Minister's Council on Climate Change.
- It hinges on the development and use of new technologies.
- The implementation of the plan includes public private partnerships and civil society action.
- The focus will be on promoting understanding of climate change, adaptation and mitigation, energy efficiency and natural resource conservation.
- There are Eight National Missions which form the core of the National Action Plan.

India’s Initiatives

National Action Plan on Climate Change

8 missions to address climate change concerns &
promote sustainable development

- National Solar Mission
- National Mission on Sustainable Habitat
- National Mission for Sustaining the Himalayan Ecosystem
- National Mission for Enhanced Energy Efficiency
- National Water Mission
- National Mission on Strategic Knowledge for Climate Change
- National Mission for a Green India
- National Mission for Sustainable Agriculture

India's Initiatives

S	National **Solar** Mission
E	National Mission for Enhanced **Energy Efficiency**
H	National Mission on Sustainable **Habitat**
W	National **Water** Mission
A	National Mission for **Sustainable Agriculture**
G	National Mission for a **Green** India
K	National Mission on Strategic **Knowledge** for Climate Change
E	National Mission for Sustaining the Himalayan **Ecosystem**
shots dekh ke **Climate Change** ho jata hai	National Action Plan on **Climate Change**

India's Initiatives

National Solar Mission (JNNSM)

- It is also known as the Jawaharlal Nehru National Solar Mission (JNNSM).
- It was inaugurated in **2010.**
- Its objectives are to establish India as a global leader in solar energy and to promote sustainable growth while addressing India's energy security challenges.
- Targets are set for three phases

1. First phase (2010-13)
2. Second phase (2013–17)
3. Third Phase (2017–22)

India’s Initiatives

- Total target of 100,000 MW (100 GW) by 2022.
- Ministry of New and Renewable Energy has proposed to achieve it through

  1. **40 GW** through Rooftop Solar Projects and
  2. **60 GW** through Large & Medium Scale solar projects.

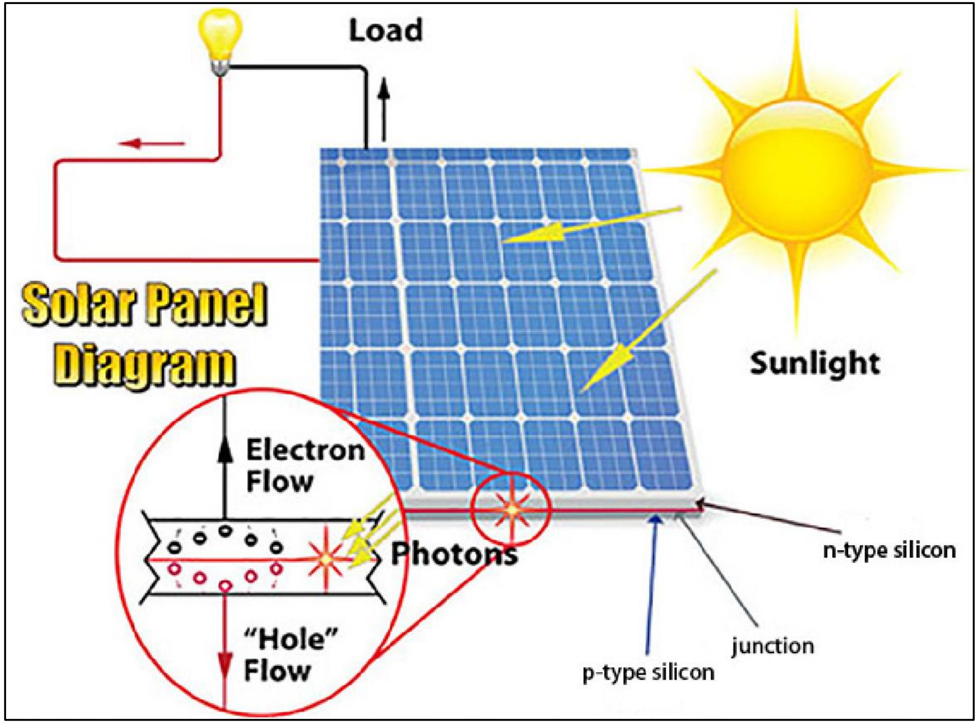
India’s Initiatives

Basics of Solar energy

- Solar energy can be converted directly into electrical energy (direct current or DC) by photovoltaic (PV) cells commonly called solar cells or to Alternating Current using Concentrating (Thermal) Systems.

Photovoltaic Technology (PV)

- **Photovoltaic cells** are made of silicon and other semiconductor materials. **(Silicon-Germanium)**
- When sunlight (photons) strikes the silicon atoms, it causes electrons to flow, creating an electrical current.
- This principle is called as the **photoelectric effect.** (Einstein was given nobel prize for this.)
- This generates DC power which is stored in Solar Battery.

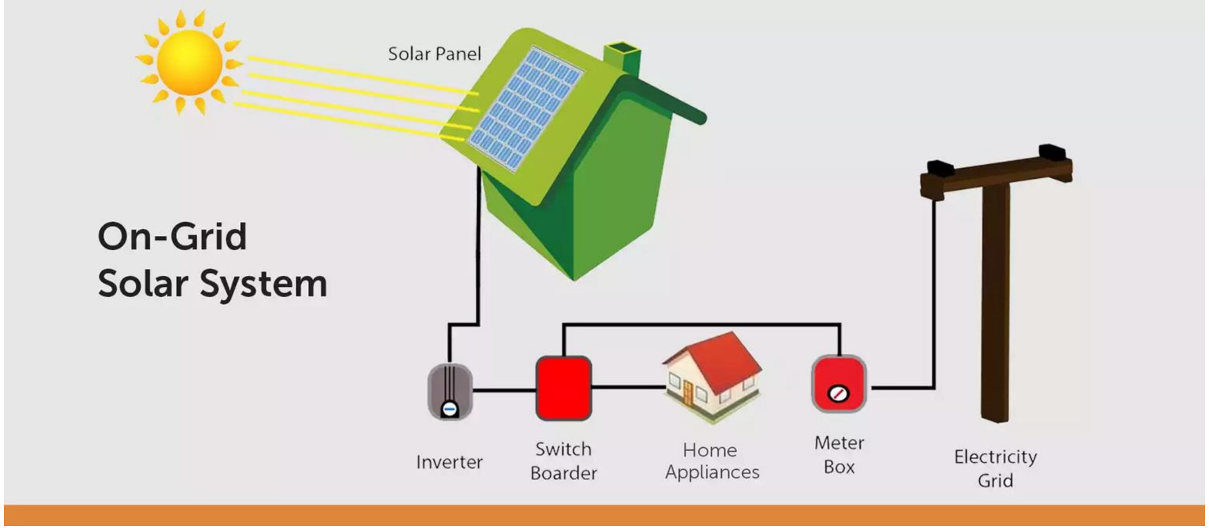

Load
Sunlight
Solar Panel Diagram
Electron Flow
Photons
"Hole" Flow
n-type silicon
junction
p-type silicon

India's Initiatives

Applications of Solar Energy can be:

a) **Off Grid:** Can be used for personal use (geyser, TV)

b) **On Grid:** Needs to be converted from DC to AC using solar inverter

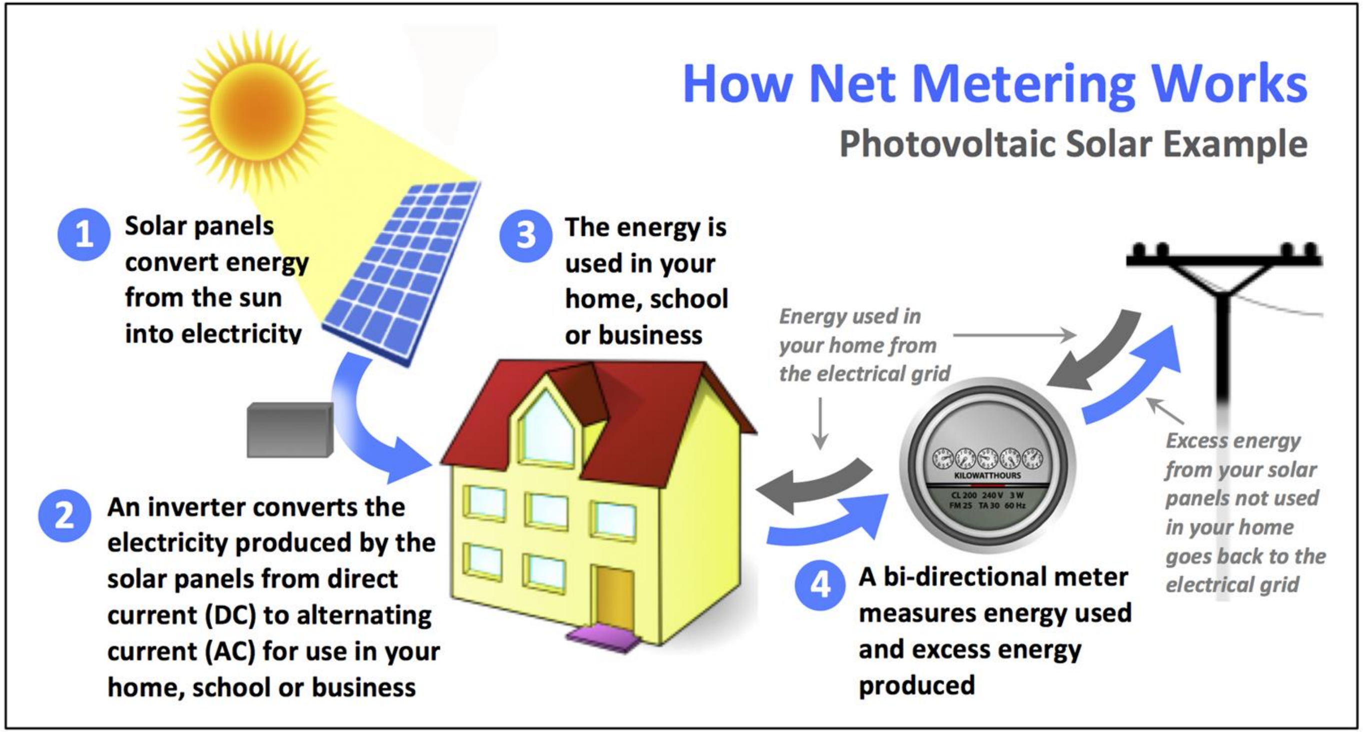


India’s Initiatives

Net metering

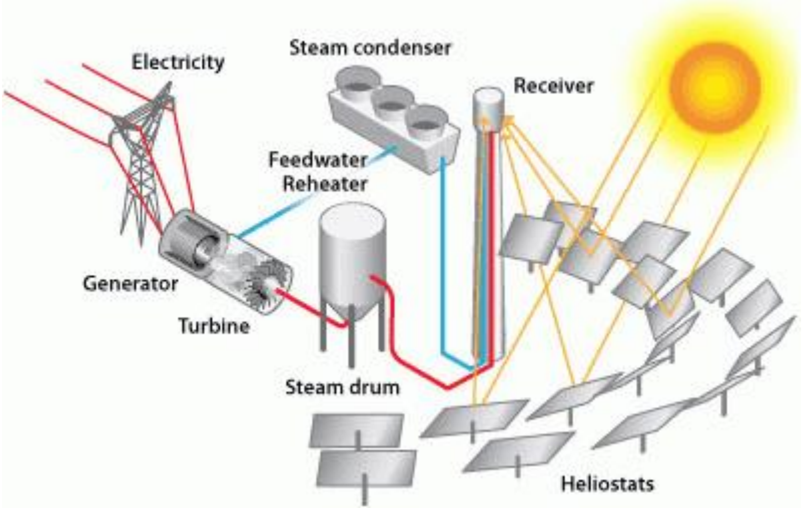
- Net Metering is a billing mechanism for grid connected Home Rooftop Solar Installation where the electricity generated by the solar panels is fed into the utility grid.
- Household draws electricity from the utility grid. The household pays **only for the difference between the energy units it consumes from the grid and the energy units fed into the grid.**
- This is measured by a bi-directional meter called Net Meter.

India’s Initiatives



Concentrating (Thermal) Systems

- Plain/curved mirror is used to concentrate solar light to generate heat. This heat is in turn used to rotate turbine and create electricity.

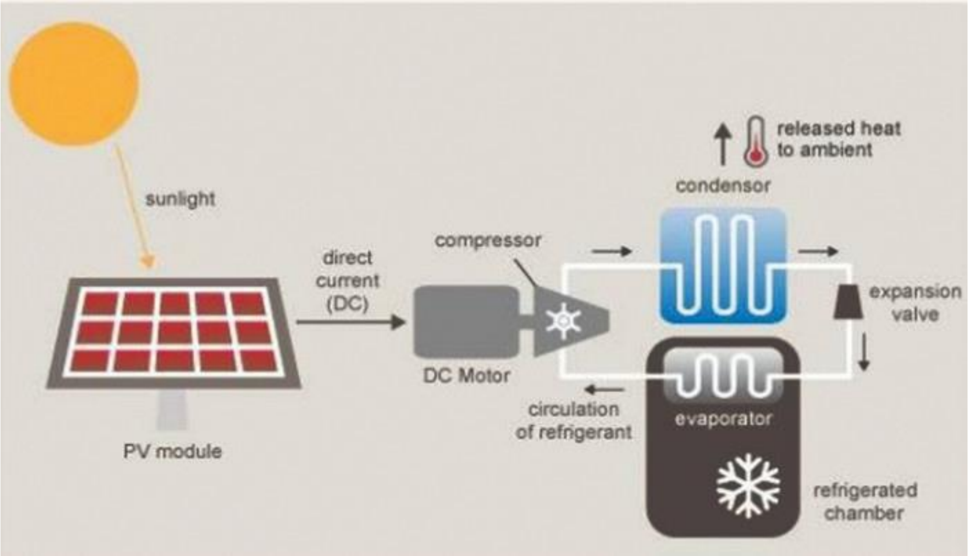


The electricity, thus produced, is AC and can be used in the grid without using inverter.

India's Initiatives

Applications of Concentrating (Thermal) Systems:

1. Boiling water.
2. Steam Generator
3. Refrigeration: **See the image given below.**

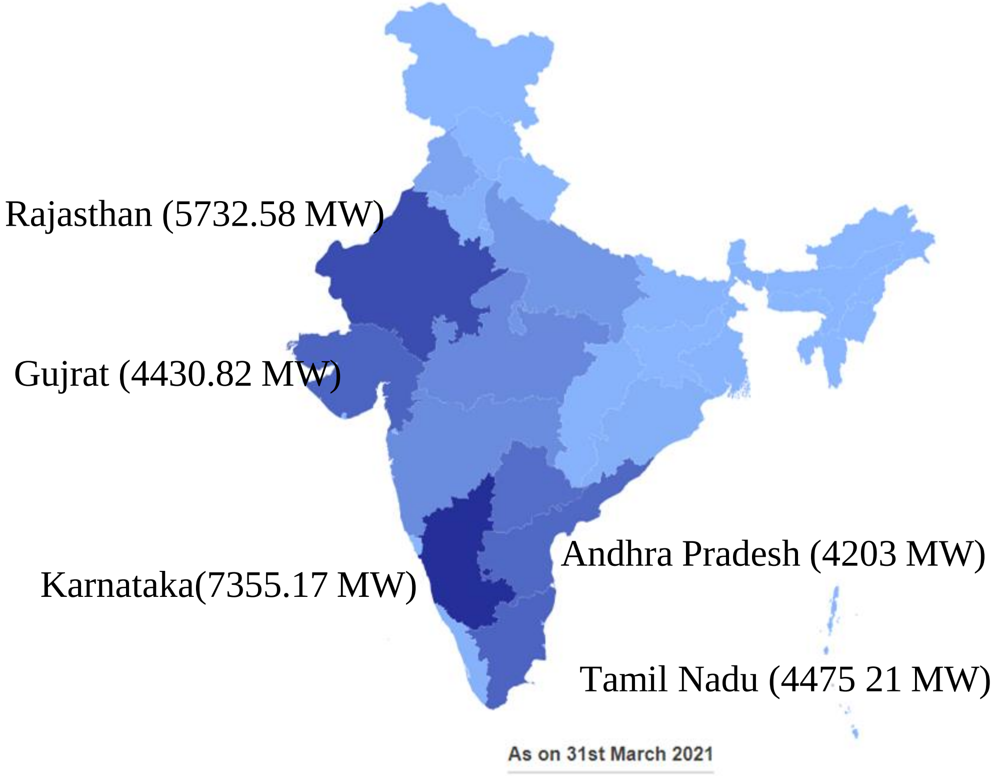


India’s Initiatives

Solar Park

- The solar park is a concentrated zone of development of solar power generation projects.
- Both PV Cells and Concentrating (Thermal) Systems are used.
- **Karnataka’s Pavagada:** Pavagada Solar Park (Tumakuru district, Karnataka) is billed as the world’s largest solar park spread over 13,000 acres of land.

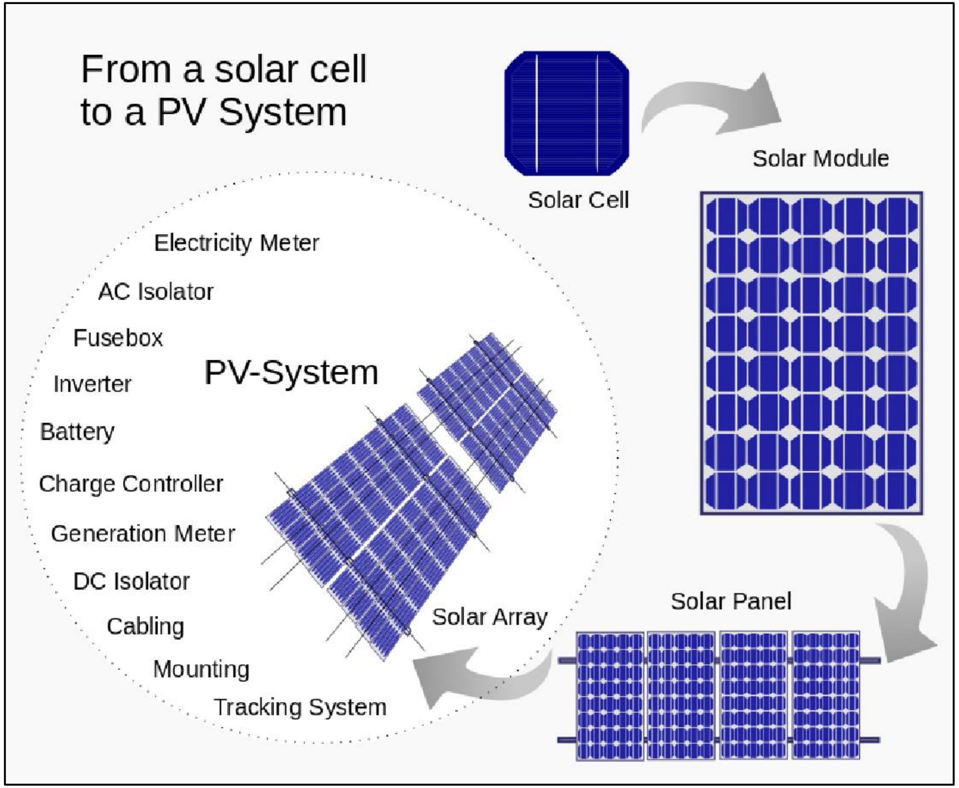
India’s Initiatives



Types of PV Modules

Crystalline Silicon (Semiconductor) PV Module	Amorphous Silicon (Semiconductor) PV Module
Costly	Suitable for low-cost product.
High Conversion Efficiency	Low Conversion Efficiency

India's Initiatives



Linkages between solar energy and agriculture

- **Current scenario**
- Two-thirds of the total irrigated area in India uses groundwater pumping, powered by more than two crore electric and 75 lakh diesel pumps.
- Farmers need reliable, affordable electricity to extract groundwater.
- Agriculture is a major consumer of electricity, accounting for one-fourth or one-third of consumption in many States.
- Currently this demand is met through subsidizing the electricity supplied to farmers as well having higher tariffs for other consumers like industry and commercial (called cross-subsidy).
- Given the criticality of agricultural sector, the electricity to farmers is supplied at low tariffs or for free.

Other Solar initiatives

- Due to the lower tariff and poor revenue collection, agricultural sales are often seen as a major reason for the financial losses of distribution companies (discoms).
- As a result, agriculture often gets poor quality supply leading to problems such as frequent pump burn-outs and power failures.

Impact of Solar energy and solar driven pumps

- **Solar PV plant**
  - A 1 MW solar plant can support around 350, 5 hp pumps and requires around 5 acres of land to set up.
  - The plant can be set up in few months and there is no change at the farmer's end.
  - Pumps need not be changed and farmers do not have to take responsibility of installation and operation.

Other Solar initiatives

- Advantages
  - Reliable day-time electricity for 8-10 hours between 8 am and 6 pm.
  - When solar generation is low, maybe due to cloud cover, balance electricity can be drawn from the discom.
  - When pumping demand is low, maybe during rains, excess solar electricity will flow back to the discom.
  - No new large transmission lines are needed, which has become a bottleneck for various large scale wind and solar power tenders.
  - Cost-effective, thereby enabling reduction in subsidy.
  - Provide distributed jobs to local youth in construction, operation and maintenance of the plant.
- It is win-win-win situation for the farmers, government and discoms, offers a much needed farmer-centric yet fiscally prudent pathway for the power sector.

India’s Initiatives

STATUS OF ADOPTION OF SOLAR ENERGY IN INDIA

- **175 GW target of renewable energy by 2022 (100 GW Solar)**
- **61 GW of solar power** had been installed
- **6.48 GW Roof Top** Solar deployment achieved (2021)
- **4th in solar photovoltaics (PV)** deployment across the globe
- **280 GW future target** for Solar Energy

Other Solar initiatives

PM KUSUM Yojana

- Pradhan Mantri Kisan Urja Suraksha evam Utthaan Mahabhiyan
- It was launched in **2019** by the **Ministry of New and Renewable Energy (MNRE).**
- State Nodal Agencies (SNAs) of MNRE will coordinate with States/UTs, Discoms and farmers for implementation of the scheme.
- **Aim:** To ensure energy security for farmers by Harvesting Solar Energy and increase the share of installed capacity of electric power from non-fossil-fuel sources to **40% by 2030** as part of Intended Nationally Determined Contributions (INDCs).
- **Target:** To add **30.8 Gigawatt (GW)** of solar capacity by FY2025-26 (Earlier target was to be completed by 2022), and **De-dieselization of Farm Sector** by replacing Diesel Pumps with Solar Pumps.

India's Initiatives

Clarity for UPSC by Dr. Shivin

Ed Sarrthi Your Own Mentor

PM-KUSUM: Components and Implementation		
Component-A: • To set up **500 KW to 2 MW** Renewable Energy based power plants by **individual farmers/ cooperatives/panchayats /farmer producer organizations (FPO),** on their barren or cultivable lands or pastureland and marshy land, referred as **Renewable Power Generator (RPG).** ○ **Projects smaller than 500 kW** may be allowed (earlier not allowed) by states based on techno-commercial feasibility. • The **power generated will be purchased by the DISCOMs** at a pre-fixed tariff determined by respective SERC. • **Performance Based Incentives** @ Rs. 0.40 per unit or Rs. 6.60 lakhs/MW/year, whichever is less, will be provided by MNRE to DISCOMs for five years for buying from RPGs.	**Component-B:** • To support 20 lakh individual farmers in installing **standalone solar pumps** of capacity up to **7.5 Horsepower (HP)** where grid supply is not available. ○ Pumps of Higher Capacity can also be installed; however, the financial support will be limited to 7.5 HP capacity. ○ It is **mandatory to use indigenously manufactured solar panels** with indigenous solar cells and modules.	**Component-C:** • **Restructured to Solarisation of agricultural feeders** instead of pumps and supports 15 lakh **individual farmers to solarise pumps** of capacity up to 7.5 HP. • For **Components B and C Centre** bears **30% of pump cost** while **70%** is borne by **State-owned DISCOMs.**

India's Initiatives

Scheme for Development of Solar Parks and Ultra Mega Solar Power Projects

- Scheme was rolled out by **Ministry for New and Renewable Energy (MNRE)** in 2014 to help solar project developers set up projects in a plug-and-play model.
- **'Plug and play'** concept normally refers to ready facilities in terms of building, power-water-sewage connectivity, road connectivity etc.
- **Target:** To set up at least 25 Solar Parks and Ultra Mega Solar Power Projects targeting over 20 GW of solar power installed capacity within a span of 5 years starting from 2014- 15.

India’s Initiatives

- Capacity of the Scheme: enhanced from 20 GW to 40 GW in 2017.
- Implementing agency: **Solar Power Park Developer (SPPD).**
- Facilitates and speeds up installation of grid connected solar power projects for electricity generation on a large scale

India’s Initiatives

India’s Largest Floating Solar Power Project

- As per **National Thermal Power Corporation,** It has commissioned India’s largest floating solar power project.
- **100 MW** Floating Solar Power Project has been operationalized at National Thermal Power Corporation- **Ramagundam, Telangana.**
- The project is endowed with advanced technology as well as environment friendly features.

India’s Initiatives

‘Gharkeupar solar is super’ campaign (Solar Rooftop)

- Minister of State for New and Renewable Energy launched the pan-India **Rooftop Solar Awareness Campaign** in a bid to mobilize public support for installing solar rooftop panels.
- The campaign is aimed at mobilizing local government, citizens, and municipalities to spread awareness of Solar Rooftop among 100 Indian towns and cities, especially **tier 2 and tier 3** towns/cities.
- Government is providing **40% of subsidy** for households to install solar panels.

Organic solar cells (OSCs)

- **Context:** Recently researchers at IIT Kanpur have developed an organic solar cell on steel substrates. It can convert a steel roof into an energy producing device.
- OSCs are one of the emerging photovoltaic (PV) technologies and are classified as third generation solar cells with organic polymer material as the light absorbing layer.

Benefits of OSCs:

- **Lightweight,** can cover a much larger area, and has **low manufacturing costs.**
- Potential to store much larger amounts of solar energies than other solar technologies.

India’s Initiatives

International Solar Alliance

- **Context:** UN General Assembly (UNGA) confers Observer Status on the International Solar Alliance (ISA)
- **Observer status of UNGA**:
    - Permanent Observers have free access to most meetings and relevant documentation
    - It started in 1946 with the **Swiss Government as first permanent observer**, a number of regional and international organizations are given observer status by UNGA.
    - Other observers include non-member states (e.g. Holy See); Intergovernmental and other organizations (e.g. ISA by resolution 76/123); and Specialized Agencies (e.g. FAO).]

India’s Initiatives

- Till October 2021, 101 countries have signed the ISA Framework Agreement and 80 countries have signed and ratified the ISA Framework Agreement.
- At the COP26 in Glasgow, US announced joining the ISA as its 101st member.
- ISA is the **first international organization** headquartered in **India**.

Initiatives taken by ISA

- **Green Grids Initiative** - One Sun, One World, One Grid (OSOWOG): It is launched by India at the global climate conference COP26 with an aim to harness solar energy wherever the Sun is shining, ensuring that generated electricity flows to areas that need it most.
- ISA **partnered with Bloomberg Philanthropies** to mobilize $1 trillion in global investments for solar energy across ISA’s member countries.

India’s Initiatives

- **<u>Global Energy Alliance for People and Planet (GEAPP)</u>** launched at COP26 with USD10 billions of committed capital to accelerate investment in green energy transitions and renewable energy solutions in developing and emerging economies.
- ISA’s **<u>Programme on Scaling Solar Applications for Agriculture Use (SSAAU)</u>** focuses on providing greater energy access and a sustainable irrigation solution to farmers through deployment of Solar Water Pumping Systems in member countries.

India’s Initiatives

Recent General Assembly meeting

- ISA approved the ‘Solar Facility’.
- **Solar Facility:** A payment guarantee mechanism expected to stimulate investments into solar projects, with two financial components:
    - **Solar Payment Guarantee Fund** to provide a partial guarantee and enable investments in geographies that do not receive investments.
    - **Solar Insurance Fund** to reduce the burden of insurance premium for solar developers in pre-revenue phase of project.

- The assembly also re-elected India and France as its President and Co-President.

India's Initiatives

Other news

- MoU between **International Solar Alliance (ISA) and International Civil Aviation Organisation (ICAO)** to check growth of CO2 emissions in the sector and idea of ICAO becoming a partner organization of ISA was mooted by India.
- Aviation sector responsible for around 2.5% of global CO_2 emissions.
- In 2015, India's **Cochin International Airport** became world's first fully Solar powered airport.

India's Initiatives

INTERNATIONAL SOLAR ALLIANCE

International Solar Alliance (ISA)

HQ
Gurugram, Haryana

Genesis: A **treaty-based intergovernmental organisation launched jointly by India and France at COP-21 in Paris in 2015** to provide a dedicated platform for cooperation among solar resource rich countries and the wider global community.

Objective: Guided by its **'Towards 1000' strategy** which aims

- to **mobilise USD 1,000 billion of investments** in solar energy solutions by 2030.
- Delivering **energy access to 1,000 million people** using clean energy solutions.
- Installation of **1,000 GW of solar energy** capacity.

Membership: 92 countries have signed and ratified the ISA Framework Agreement.

Member

Other key information:

- UN General Assembly conferred **Observer Status to the ISA in 2021.**
- The **ISA assembly is the apex decision-making body** of the 110-members.
- It is **open for all member states.**
- **Major Initiatives under ISA**
  - **ISA Solar Technology and Application Resource Centre (ISTAR C)** to support capacity building efforts in the ISA member countries through training.
  - **ISA Solar Fellowship for Mid-Career Professionals,** for the creation of a skilled and qualified professional manpower for management of solar energy projects etc.
  - **One Sun One World One Grid (OSOWOG),** building a global ecosystem of interconnected renewable energy resources, launched in **partnership with ISA and World Bank Group.**

India’s Initiatives

National Mission for Enhanced Energy Efficiency

- Objective: Promote the market for energy efficiency by fostering innovative policies & market instruments.
- The NMEEE mission document, which was approved in **2010**, established the immense energy efficiency potential of India, which was about **Rs. 74,000 crores.**
- A recent World Bank study has estimated the country's energy efficiency market to be at **1.6 lakh crores**.
- NMEEE includes 4 efficiency initiatives under its umbrella

1) Perform Achieve and Trade (PAT)

- Assigning energy reduction targets to large energy intensive industries and distributing **Energy Saving Certificates (ESCerts)** on achievement of the targets. These ESCerts can then be traded. Consumers who are not able to meet their energy savings targets will buy the ESCerts.

India’s Initiatives

2) Market Transformation for Energy Efficiency (MTEE)

- Promoting adoption of energy efficient equipment and appliances through innovative business models.
- Programs that were developed under this scheme include:
  - **Domestic Efficient Lighting Program:** Unnat Jeevan by Affordable LEDs for All **(UJALA) program** to promote the use of LED lighting for households.
  - **Super-Efficient Equipment Program (SEEP):** Under this program, the manufacturers are incentivized by GOI to elevate the efficiency standards of the equipment.
  - **Bureau of Energy Efficiency** (Ministry of Power) launched the program in the XII five-year plan with a focus on ceiling fans, considering its wide use and impact on domestic energy consumption.

3) Energy Efficiency Financing Platform (EEFP)

- The EEFP initiative is intended towards catalysing the **finances** for energy efficiency sector by addressing the barriers and challenges in market development and project implementation.
- It provides a **platform** for financial institutions, investors and project developers to increase their confidence in supporting energy conservation and efficiency projects.

4) Framework for Energy Efficiency Economic Development (FEEED)

- Promoting energy efficiency initiatives by hedging against investment risks.
- BEE institutionalized two types of funds in order to protect the confidence of banks and investors in energy efficiency projects and to avoid the stalling of projects:
  1. **Partial Risk Guarantee Fund for Energy Efficiency (PRGFEE):** The fund guarantees a risk cover for banks and investors for up to 50% loan amount or INR 10 crore per project, whichever is less.
  2. **Venture Capital Fund for Energy Efficiency (VCFEE):** This fund is intended towards promoting equity financing (stock, share) in the energy efficiency sector and thus reducing the impact of non-availability of debt financing (bond, loan) to small size companies and projects. The equity support is equivalent to INR 2 crore or 15% of total equity whichever is less

India’s Initiatives

Recent Energy Efficiency Initiatives

- **<u>Energy Conservation (Amendment) Act</u>**
- **Context-** It came into force recently
- **Background**
    - Amends Energy Conservation Act, 2001 which provides a framework for regulating energy consumption and promoting energy efficiency and energy conservation.
    - While 2001 act deals with saving energy, 2022 amendment deals with saving the environment and tackling climate change, thus broadening scope and objective of principal Act.

India's Initiatives

Key Features:

- Empowers central government to specify a **carbon credit trading scheme**.
    - Carbon credit implies a tradable permit to produce a specified amount of CO_2 or other GHGs. (Recall our earlier discussion on Kyoto Protocol)
    - Central government or any authorized agency may issue **carbon credit certificates** to entities registered and compliant with scheme.
- Government is empowered to set requirements for designated consumers to meet a **minimum share of energy consumption** from non-fossil sources like green hydrogen, green ammonia, etc.
    - Failure to meet obligation will be **punishable with a penalty of up to Rs 10 lakh.**

India’s Initiatives

- **‘Energy Conservation and Sustainable Building Code’** to replace Energy Conservation Code for buildings.
  - This new code will provide norms for energy efficiency and conservation, use of renewable energy, and other requirements for green buildings.
  - Also applicable to the **office and residential buildings** meeting above criteria. It also empowers state governments to lower the load thresholds.
- **Expands** the scope of energy consumption standards to include **vehicles** (as defined under the Motor Vehicles Act, 1988), and **vessels** (includes ships and boats), in addition to equipment and appliances.
- **State Electricity Regulatory Commissions (SERCs)** are empowered to make regulations for discharging their functions.

India’s Initiatives

- State Governments required to constitute **energy conservation funds** for promotion of energy efficiency and conservation measures. It will receive contribution by both Union and State government.
- **Increases** and **diversifies** number of members and secretaries in **governing council of BEE.**
  - The Government of India set up Bureau of Energy Efficiency (BEE) on 1st March 2002 under the provisions of the Energy Conservation Act, 2001.

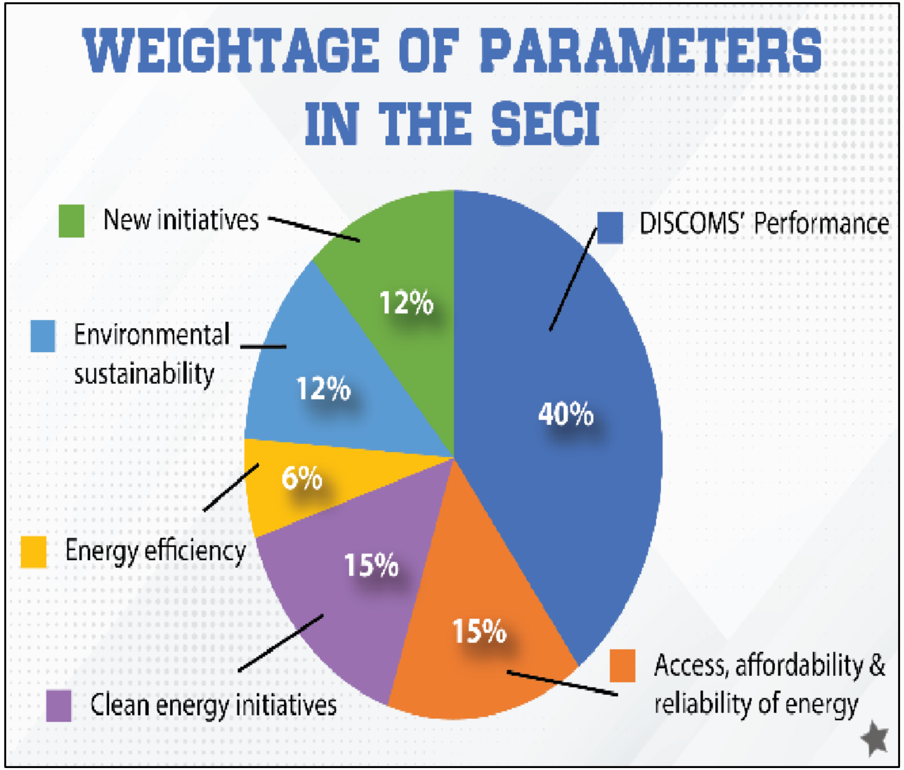
State Energy & Climate Index (SECI)-Round I

- Released by **NITI Aayog recently**
- It is the first index that aims to track the efforts made by states and union territories (UTs) in the climate and energy sector.

Objective of the ranking

- Ranking the States based on their efforts towards improving energy access, energy consumption, energy efficiency, and safeguarding environment;
- Encouraging healthy competition among the states on different dimensions of energy and climate

India’s Initiatives



CATEGORY	SECI SCORE
Front-runners (Top one-third)	Composite SECI score>=46
Achievers (Middle one-third)	Composite SECI score one-third) between 36 and 46
Aspirants (Lowest one-third)	Composite SECI score<=36

Performance of the states and UTs:

- More than half the states **scored higher than the average.**
- Overall performance- Top Scorer- **Chandigarh,** Lowest scorer- **Lakshadweep**
- Top 3 performers based on classification
  - Larger States: **Gujarat,** Kerala, and Punjab.
  - Smaller States: **Goa**, Tripura, and Manipur.
  - Union Territories: Chandigarh, Delhi, and Daman & Diu/Dadra & Nagar Haveli.

India’s Initiatives

Other Energy Indexes across the globe and India's performance

Index	World Energy Trilemma Index(WETI)	Energy Transition Index (ETI)	Renewable Energy Country Attractiveness Index (RECAI)	Climate Change Performance Index (CCPI)
Publishing Agency	World Energy Council	World Economic Forum (WEF)	Ernst & Young (EY)	German watch eV.
What it measures	Measures energy system performance in terms of Energy Security, Ener Equity, Environmental Sustainability in Country context	Checks nation's energy system information	Ranks performance of economies based on the investment made in the renewable energy sector - energy supply, renewable technologies, & ease of doing business	Measures country's progress towards the NIDC 2030 targets and compares climate protection performance of countries
India’s Rank	75/127(2021)	87/115(2021)	3/40(2021)	3/40(2021)
Best performing countries	Top 3: Sweden, Switzerland, Denmark	Top 3: Sweden,Norway, Denmark	Top 2: USA & Mainland China	Top 6: Denmark (4^{th}), Sweden (5^{th}), Norway (6^{th})

Star Labelling Programme (BEE Star Ratings)

- The Ministry of Power had launched the Standards and Labeling (S&L) Scheme under the **Energy Conservation Act, 2001.**
- The **Bureau of Energy Efficiency (BEE)** undertook the responsibility of the scheme.
- The key objective of the scheme is to provide consumers an informed choice about the energy savings of high energy equipment and appliances.
- To materialise the objective of the scheme, the famous **‘star label’** was introduced on appliances – that detailed the efficiency of the product along with various particulars, allowing consumers to compare models and choose the best one in terms of energy conservancy.

India’s Initiatives

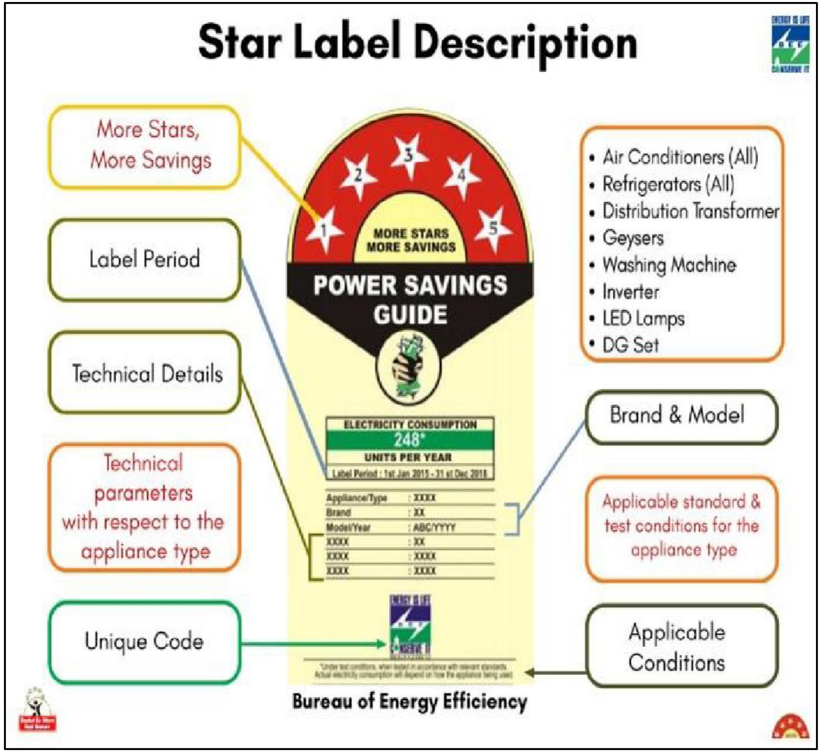
- Presently, Star Labelling program covers star ratings for about 29 appliances/ equipment and it is mandatory for the following appliances:

Mandatory Appliances	Voluntary Appliances
Room Air Conditioners	Induction Motors
Frost Free Refrigerators	Agricultural Pump Sets
Tubular Florescent Lamp	Ceiling Fans
Distribution Transformer	Domestic Liquefied Petroleum Gas (LPG) Stoves
Room Air Conditioner (Casettes, Floor Standing Tower, Ceiling, Corner AC)	Washing Machine
Direct Cool Refrigerator	Computer (Notebook /Laptops)

India's Initiatives

Mandatory Appliances	Voluntary Appliances
Color TV	Ballast (Electronic/Magnetic)
Electric Geysers	Office equipment's (Printer, Copier, Scanner, MFD's)
Inverter Air conditioners	Diesel Engine Driven Monoset Pumps for Agriculture
LED Lamps	Solid State Inverter
	Diesel Generator
	Chillers
	Microwave Ovens

India’s Initiatives



India's Initiatives

Initiative	Details

Promoting energy efficiency and renewable energy in selected MSME clusters in India project	• Project's success was showcased in National Conclave which was organised by Bureau of Energy Efficiency (BEE).
	• Project was initiated in 2011 and **executed by United Nations Industrial Development Organization (UNIDO) in collaboration with BEE.**
	  o Funded by **Global Environment Facility (GEF).**
	  o Aims to **develop market environment for introducing energy efficiency** and enhance use of renewable energy in selected energy-intensive MSME sector.
	• Has been **supported by Ministry of Micro, Small, and Medium Enterprises** and **Ministry of New and Renewable Energy.**

Saksham 2022	• Sakshan 2022 was launched recently.
	• **An annual one-month long people centric fuel conservation campaign of the Petroleum Conservation Research Association (PCRA).**
	  o **PCRA** is a registered society set up under the aegis of **Ministry of Petroleum & Natural Gas.**
	  o It helps the government in **proposing policies and strategies for petroleum conservation,** aimed at reducing excessive dependence of the country on oil requirement.
	  o **Aim:** To **spread the message of fuel conservation and greener environment across India.**

National Mission on Sustainable Habitat

- The National Mission on Sustainable Habitat was approved in 2010.
- It seeks to promote:
    - Improvements in **energy efficiency** in buildings by extending energy conservation building code to new and large commercial buildings.
    - Better **urban planning** and efficient and convenient public transport to facilitate the growth of medium and small cities.
    - Improved management of **solid and liquid waste**, e.g., recycling of material and urban waste management.
    - Improved ability of habitats to adapt to climate change and measures for improving advance warning systems for extreme weather events.
    - Conservation through appropriate changes in legal and regulatory framework.

India’s Initiatives

- The Mission is being implemented through the following programmes of **Ministry of Housing and Urban Affairs:**

  1) Atal Mission on Rejuvenation and Urban Transformation (AMRUT)
  2) Swachh Bharat Mission
  3) Smart Cities Mission
  4) Urban Transport Programme

Other Sustainable Habitat initiatives in news:

Sustainable cities integrated approach pilot (SCIAP) project

- **Context:** UN-Habitat has highlighted issues related to Jaipur's urban development to propose strategic interventions and promote sustainable development.

SCIAP project

- Implemented by: **UNIDO and UN-Habitat** in partnership with the **Ministry of Housing and Urban Affairs (MoHUA),** Government of India.
- Funded by: **Global Environment Facility**
- Covers 5 pilot cities: **Jaipur, Mysore, Vijayawada, Guntur and Bhopal.**

India's Initiatives

- **Goal:** To infuse **sustainability strategies into urban planning** and management at the city level and create an enabling climate for investments in green infrastructure that would reduce GHG emissions, improve service delivery and enhance the quality of living for all citizens, thereby building resilience and strengthening the governance capacity of the cities.
- Major component of the project: To develop an **Urban Sustainability Assessment Framework (USAF)** for spatial planning in India.
- It is designed as a decision support tool for municipal commissioners and urban practitioners to support sustainable and resilient urban planning and management of cities in India.
- Jaipur has received an overall **sustainability rating of three** on USAF.

India's Initiatives

United Nations Human Settlements Programme (UN-Habitat)

UN-HABITAT

HQ
Nairobi, Kenya

Genesis: **UN's agency for human settlements,** mandated by UN General Assembly

Objective: To **promote socially and environmentally sustainable towns and cities** to provide adequate shelter for all.

Other key information: UN-Habitat Publications

- The State of the World's Cities
- The Global Report on Human Settlements
- New Urban Agenda
- Sustainable Development Goals and Urban
- Local Bodies – The Future We Want

India’s Initiatives

11th World Urban Forum (WUF), 2022

- Held at **Katowice**, Poland, the WUF is the premier global conference on sustainable urbanization co-organized by **UN-Habitat.**
- WUF was established in **2001** by the UN to examine rapid urbanization and its impact on communities, cities, economies, climate change and policies.
- First WUF was held in **Nairobi, Kenya** in 2002.
- At 11th WUF, the National Institute of Urban Affairs (NIUA) Climate Centre for Cities (NIUA CCube), World Resources Institute India (WRI India) etc. launched India’s **first national coalition platform** for urban **nature-based solutions (NbS).**

India's Initiatives

GRIHA (Green Rating for Integrated Habitat Assessment)

- GRIHA was developed by **TERI** (private energy research institute) and was adopted as national rating system for green buildings by Government of India in 2007.
- It evaluates **environmental performance** of a **building** holistically over its entire life cycle, thereby providing a definitive standard for what constitutes a **'green building'**.
- It is also recognized as India's own green building rating system in INDIA's INDC submitted to UNFCCC

India’s Initiatives

CITIIS (City Investments to Innovate, Integrate & Sustain) program

- CITIIS is a joint program of the **Ministry of Housing and Urban Affairs, the French Development Agency the European Union, and NIUA.**
- Aim: to provide **financial assistance** by way of grants and technical assistance through international and domestic experts.
- Main component of the ‘Program to **fund Smart City** projects through a Challenge Process.’

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National Mission for Sustaining the Himalayan Ecosystem (NMSHE)

- The primary objective of the Mission is to:
  - develop a **sustainable model** to continuously assess the health status of the Himalayan Ecosystem, and
  - enable **policy bodies in policy-formulation** as also to assist States in the Indian Himalayan Region with implementation of actions for sustainable development.

India’s Initiatives

- The NMSHE attempts to address a variety of issues:
  - **Himalayan glaciers** and associated consequences.
  - Prediction and management of natural hazards.
  - Biodiversity conservation and protection.
  - **Wildlife conservation** and protection.
  - **Traditional knowledge** societies and their livelihood.

India’s Initiatives

- **<u>The effect of Climate Change on Himalayan glaciers and associated hydrological consequences:</u>**
  - Increased drought like situations due to overall decrease in the number of rainy days.
  - Increased flood events due to overall increase in the rainy day intensity.
  - Effect on groundwater quality in alluvial aquifers due to increased flood and drought events.
  - Influence on groundwater recharge due to changes in precipitation and evaporation.
  - Increased saline intrusion of coastal and island aquifers due to rising sea levels.

National Water Mission

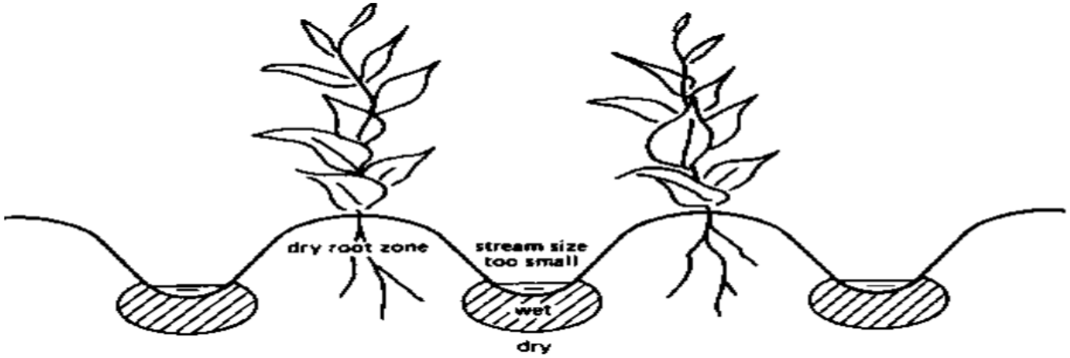
The Mission is in line with National Water Policy which aims to:

- increase **water use efficiency by 20%**,
- ensure that a considerable share of the water needs of urban areas are met through recycling.
- ensure that the water requirements of coastal cities are met through modern desalination technologies.
- ensure **basin level management strategies** by working with states to deal with variability in rainfall.

National Water Mission

The Mission aims to achieve its objectives through:

- Increasing efficiency through regulatory mechanisms (differential entitlements and pricing).
- Enhanced storage both above and below ground, rainwater harvesting.
- Incentivising water-neutral or water-positive technologies, and adoption of large scale irrigation programmes which rely on sprinklers, drip irrigation and **ridge and furrow irrigation** (The crops are grown on the ridges and the furrows are used to irrigate.)



Initiatives

Dynamic Ground Water (GW) Resource Assessment 2022 Report

- It is released by Ministry of Jal Shakti
- It is carried out at **periodical intervals** jointly by **Central Ground Water Board and States/UTs.**

Key highlights:

- India is largest user of GW with a 1/4th of total global withdrawal.
- Total annual **GW recharge** has increased and **rainfall** contributes to nearly **61 %** in this.
- State of GW extraction (percentage utilisation vs recharge) saw a decline reaching at 60%

Dynamic Ground Water (GW) Resource Assessment 2022 Report

Key highlights:

- **Categories:**
    - 67% GW units are safe. (Ground water extraction is **less than 70%**.)
    - There was a decline in the number of over-exploited units. 14 % assessment units are 'Overexploited' (Ground water extraction **exceeding the annually replenishable ground water recharge**.)
    - 4% are 'Critical'. (Ground water extraction is between **90-100** % of annual tractable resources available.)

Initiatives

- **Central Ground Water Board (CGWB) is the apex organization of the Ministry of Water Resources dealing with ground water and related issues.**

- **The Union Government has constituted the Central Ground Water Authority (CGWA) on 14th January 1997 under Section 3(3) of Environment (Protection) Act, 1996 with objective to regulate and control development and management of ground water with jurisdiction in whole of the country.**

UN Water Summit on Ground Water

United Nations (UN) Water Summit on Groundwater (GW) 2022

- Organized in **Paris** by **UN-Water, UNESCO and International Groundwater Resources Assessment Centre (IGRAC)**
- To bring attention to groundwater at highest international level.
- Will mark the completion of **"Groundwater: Making the invisible visible" campaign** run by **UN-Water** throughout **2022.**

- **UN-Water:** A **UN inter-agency coordination mechanism** for all freshwater and sanitation related matters.
- **IGRAC:** A **UNESCO Centre working under World Meteorological Organisation (WMO),** and financially supported by **Netherlands.** It specializes in regional and transboundary-level assessment and monitoring of GW resources.

United Nations (UN) Water Summit on Groundwater (GW) 2022

- UN water development report 2022 will be used as a baseline for the global acceleration framework on SDG - 6.

Water Convention

Water Convention (Convention on the Protection and Use of Transboundary Watercourses and International Lakes 1992)

- An international legal instrument steered by **United Nations Economic Commission for Europe (UNECE)** and intergovernmental platform.
- It is adopted in **Helsinki in 1992** and entered into force in 1996.
- Initially negotiated as a regional instrument, opened globally for accession to all UN Member States in **2016.**
- Requires Parties to use transboundary waters in a reasonable and equitable way and ensure their sustainable management.
- Parties bordering the same transboundary waters must cooperate by entering into specific agreements and establishing joint bodies.

Water Convention

- The United Nations Economic Commission for Europe (UNECE) was set up in **1947 by the United Nations Economic and Social Council (ECOSOC)** to promote pan-European economic integration.
- Remember if you have seen IR lectures : does india have an important water sharing treaty with a significant neighbor? What is the dispute settlement mechanism?
- India is NOT part of water convention, NOT part of UNECE.

Water Convention

World Water Forum

- In 2022, Water Convention organized the first ever transboundary pavilion at the ninth World Water Forum in **Dakar, Senegal.**
- World's largest event on water, organized every three years since **1997 by the World Water Council** (a think tank), in partnership with a host country.
- It provides a unique platform where the international water community and key decision makers can collaborate on global water challenges

Unconventional Water Resources

- Commonly include waters of inferior or marginal quality like- saline water, brackish water, agricultural drainage water, treated or untreated wastewater effluents etc.

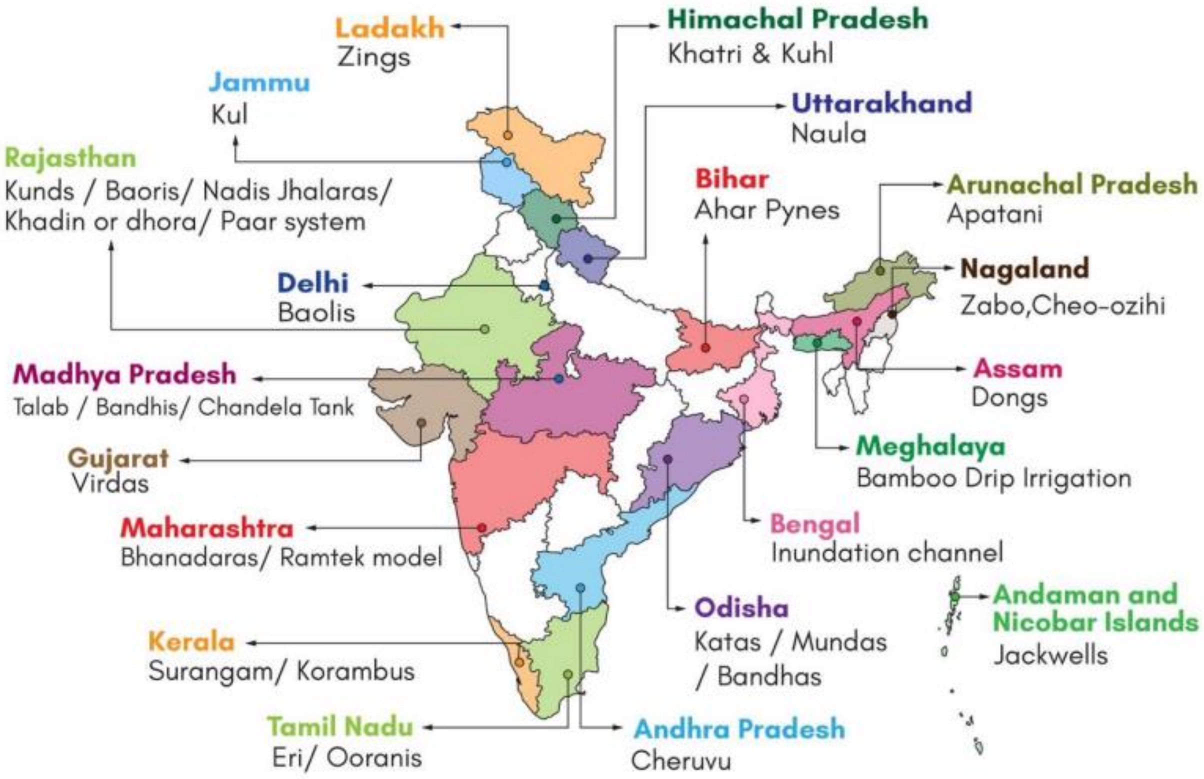
- Use of this water requires adoption of more complex management practices and stringent monitoring procedures.

Unconventional Water Resources

6 categories of UWRs

- **Harvesting water from air and ground** by Cloud Seeding or Rain Enhancement, Fog Harvesting.
- **Fog harvesting and micro-catchment rainwater harvesting** marked as low-cost and low-impact methods.
- **Desalination**- Removing salts or other minerals from wastewater effluent, to obtain fresh water for human consumption.
- **Tapping fresh and brackish groundwater** offshore and onshore
- **Reusing used water** i.e. municipal waste-water and agricultural drainage water.
- **Moving water physically** to water-scarce areas in ships' ballast holds, or towing icebergs.
- **Micro-scale capture of rainwater** that would otherwise evaporate.

Unconventional Water Resources



Grey Water

Grey Water

- **Context:** 100% Saturation of Grey Water Management has been achieved in **Pappankuzhi Village, Tamil Nadu.**

- Grey water refers to **wastewater from baths, showers, hand basins, washing machines, dishwashers and kitchen sinks, excludes streams from toilets.**

- **Significance of grey water recycling:**
    - Prevent potential harm to the environment and reduce the demand for freshwater.
    - Reliable water resource unlike rainwater harvesting.
    - Good fertilizer source due to high nitrogen and phosphorus content.

Bharat Tap

- The Ministry of Housing and Urban Affairs (MoHUA) has launched Bharat Tap initiative.
- It is conducted under aegis of **Atal Mission for Rejuvenation and Transformation 2.0 (AMRUT)** and **Swachh Bharat Mission 2.0 (SBM).**
- **Aim:** To provide low-flow, sanitary-ware at scale, and thereby reduce water consumption at source considerably.
- It is estimated to save minimum 40% water, in turn result into energy saving.

Other Initiatives

Nirmal Jal Prayas

- MoHUA launched ‘Nirmal Jal Prayas’, initiative of National Real Estate Development Council (NAREDCO) Mahi.
- It aims to map groundwater and save **500 crore litres** of water per annum.
- Through the initiative, advocacy, awareness and amplification towards saving water will be disseminated and highlighted.

NAREDCO

- It is an autonomous self-regulatory body, established in **1998**, under aegis of MoHUA.
- It strives to be the collective force influencing and shaping real estate industry.
- NAREDCO had established **Mahi - NAREDCO Women's Wing for empowering women entrepreneurs** and encouraging participation of women in real estate sector and allied fields.

Other Initiatives

National Water Awards (NWA)

- Ministry of Jal Shakti has launched 4th NWA on Rashtriya Puraskar portal.
- This award is introduced to recognize and encourage exemplary work and efforts made by States,
- Districts, individuals, etc. in accomplishing government's vision **'Jal Samridh Bharat'.**
- Award winners in different categories will be presented with a citation, trophy, and cash prize.

JALDOOT App

- It has been developed by **Ministry of Rural Development.**
- It aims to identify the ground water level in selected villages.
- **Gram Rojgar Sahayak (GRS)** will measure the water level of selected wells twice a year (pre monsoon and post-monsoon).
- Data collected could be utilised as part of the Gram Panchayat Development Plan (GPDP) and Mahatma Gandhi NREGA planning exercises.

Other Initiatives

Swachh Sujal Pradesh

- **Andaman and Nicobar (A&N) Islands** have become India's first Swachh Sujal Pradesh.
- Swachh Sujal Pradesh certification is provided by **Ministry of Jal Shakti.**

Three Components:

- Safe and secure drinking water supply and management
- ODF (open defecation free) Plus: ODF Sustainability and Solid and Liquid Waste Management (SLWM), and
- Cross-cutting interventions like convergence, IEC (Information, Education Communication), action planning, etc.

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Pey Jal Survekshan

- **Ministry of Housing and Urban Affairs** conducted ground survey of PJS under **Atal Mission for Rejuvenation and Urban Transformation (AMRUT) 2.0.**
- It serves as a monitoring tool and an accelerator for AMRUT Mission.

Sponge City

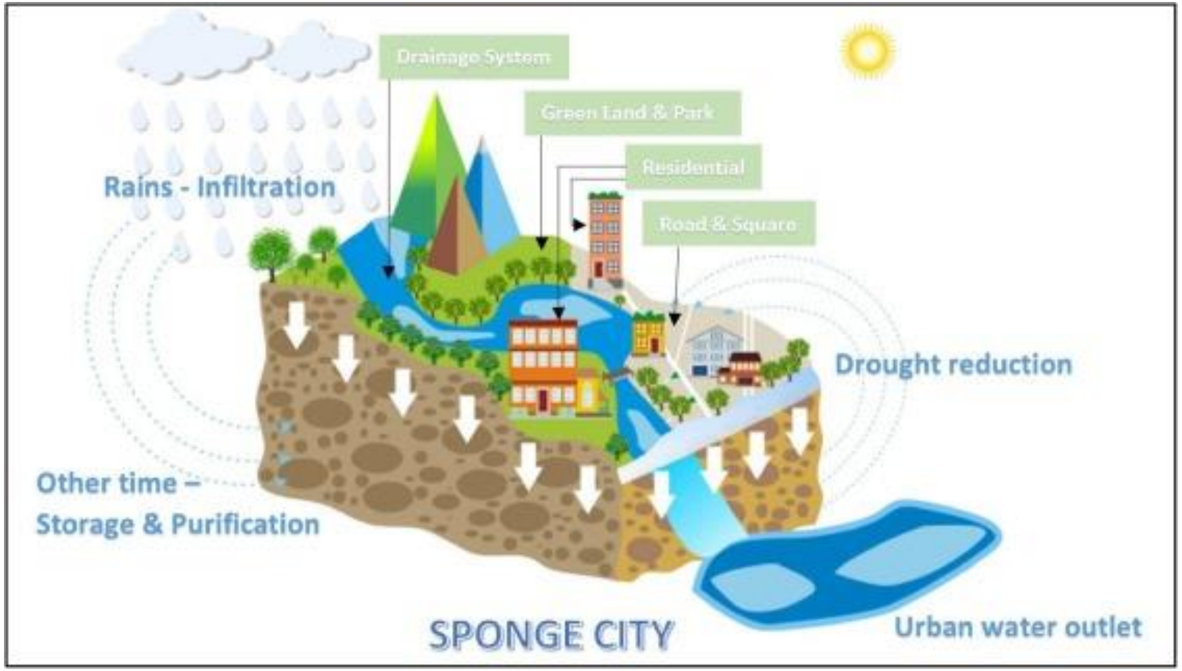
Sponge City

- Recently, Urban flooding seen in Auckland and sponge city concept can avoid such future disasters.

- **Sponge city** is a city that is designed to passively absorb, clean, and use rainfall in an ecologically friendly way that reduces dangerous and polluted runoff. It incorporates green roofs, rain gardens, and permeable pavements to absorb and filter water.

- In early 2000s, Chinese architect **Kongjian Yu** created the concept of "sponge city".

Sponge City


Drainage System
Green Land & Park
Residential
Road & Square
Rains - Infiltration
Drought reduction
Other time –
Storage & Purification
SPONGE CITY
Urban water outlet

Other Initiatives

River Cities Alliance (RCA)

- **DHARA 2023 (Driving Holistic Action for Urban Rivers),** annual meeting of RCA members, was held. DHARA provides a platform to co-learn and discuss solutions for managing local water resources.

- **River Cities Alliance** is a dedicated platform for river cities to ideate, discuss and exchange information for sustainable management of urban rivers.
- It includes cities from both **Ganga basin and non- Ganga basin states.**
- RCA is a successful partnership of **Ministry of Jal Shakti and Ministry of Housing and Urban Affairs.**

National Mission for Clean Ganga (NMCG)

- It was registered as a society in 2011 under the Societies Registration Act 1860.
- It acted as the implementation arm of National Ganga River Basin Authority (NGRBA) which was constituted under the provisions of the Environment (Protection) Act (EPA),1986.
- NGRBA has since been dissolved with effect from 2016 consequent to the constitution of National Council for Rejuvenation, Protection and Management of River Ganga which is referred as **National Ganga Council.**

National Mission for Clean Ganga (NMCG)

Aim & Objective of NMCG

- To ensure effective abatement of pollution and rejuvenation of the river Ganga by adopting a river basin approach to promote inter-sectoral coordination for comprehensive planning and management.
- To maintain **minimum ecological flows** in the river Ganga with the aim of ensuring water quality and environmentally sustainable development.

National Mission for Clean Ganga (NMCG)

Five tier structure:

- **National Ganga Council** under the chairmanship of Prime Minister of India.
- **Empowered Task Force (ETF) on river Ganga under** chairmanship of Union Minister of Jal Shakti (Department of Water Resources, River Development and Ganga Rejuvenation).
- **National Mission for Clean Ganga (NMCG).**
- **State Ganga Committees.**
- **District Ganga Committees** in every specified district abutting river Ganga and its tributaries in the states.

District Ganga Committees (DGCs)

- **In News:** Minister for Jal Shakti launched **Digital Dashboard for DGCs Performance Monitoring System.**

- DGCs are constituted in districts on Ganga River basin to ensure people's participation in management and pollution abatement in Ganga and its tributaries.
- District Collector is the Chairperson of DGC.

National Mission for Clean Ganga (NMCG)

NMCG has a **two-tier management structure** and comprises of:

- Governing Council
- Executive Committee
- Both are headed by Director General, NMCG.
- The Executive Committee has been authorised to accord approval for all projects up to Rs.1000 crore.
- State Programme Management Groups (SPMGs) acts as the implementing arm of State Ganga Committees.

Ganga Praharis are motivated and trained volunteers from among the local communities working for biodiversity conservation and cleanliness of the Ganga River with the ultimate objectives of restoring the **Nirmal (Unpolluted Flow) and Aviral Dhara (Continuous Flow).**

National Mission for Clean Ganga (NMCG)

Namami Gange Programme

- Namami Gange Programme is an **Integrated Conservation Mission,** approved as a ‘Flagship Programme’ by the Union Government in June 2014 to accomplish the twin objectives of effective abatement of pollution and conservation and rejuvenation of National River Ganga.
- It is being operated under the **Department of Water Resources, River Development and Ganga Rejuvenation, Ministry of Jal Shakti.**
- The program is being implemented by the National Mission for Clean Ganga (NMCG), and its state counterpart organizations i.e State Program Management Groups (SPMGs).

National Mission for Clean Ganga (NMCG)

The main pillars of the programme are:

- Sewage Treatment Infrastructure
- River-Front Development
- River-Surface Cleaning
- Biodiversity
- Afforestation
- Public Awareness
- Industrial Effluent Monitoring
- Ganga Gram

Arth Ganga

- Arth Ganga is a concept espoused by the Prime Minister during National Ganga Council meeting in Kanpur in 2019.
- It focuses on creating **economic livelihood opportunities** to sustain the activities under Namami Gange Programme, the flagship program of the Government to clean Ganga and its tributaries.
- **Aim:** To contribute about 3% to the GDP from Ganga Basin. It expects to generate economic benefit of more than Rs 1000 crores over the next 5 years.

Arth Ganga

Institutional Building
• By enhancing the local capacities for better decentralized water governance.
Zero Budget Natural Farming
• Chemical-free farming on 10km on either side of the river.
• Generating "more income perdrop".
• 'Gobar Dhan' for farmers.
Culture Heritage & Tourism
• Introduction of boat tourism through community jettis, promotion of yoga, adventure tourism etc. and Ganga Artis.
Verticals of Arth Ganga
Monetization and Reuse of Sludge & Wastewater
• Reuse of treated wastewate in irrigation, industries and revenue generation for ULBs.
Public Participation
• To ensure Increased synergies between stakeholders.
Livelihood Generation Opportunities
• Ghat Mein Haat
• Promotion of local products, Ayurveda, medicinal plants etc.
• Capacity building of volunteers like Ganga Praharis.

Arth Ganga

New initiatives launched under Arth Ganga

Jalaj
- Launched at 26 locations on Ganga basin main stem states.
- Involves setting up of **small shops or floating mobile centres.**
- Aims to **promote livelihood** on the banks of River Ganga.

MoU between NMCG and Sahakar Bharti
- To identify 75 villages in 5 states on the main stem as **'Sahakar Ganga Grams'.**
- Aims to promote **natural farming** among the farmers, FPOs and Cooperatives and **facilitating marketing** of natural farming/organic produce under **brand Ganga.**

Tourism-related portal ImAvatar
- For facilitating **tourism, marketing of local products** (both agriculture and handicrafts), and **sustainability of ghats** and other assets created by NMCG.

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Stockholm World Water Week 2022

- The World Water Week is an annual event organized by Stockholm International Water Institute (SIWI) to address the global water issues and related concerns of international development.
- **SIWI:** A not-for-profit institute with a wide range of expertise in water governance – from sanitation and water resources management to water diplomacy.

National Mission for Sustainable Agriculture

- NMSA has been formulated for enhancing agricultural productivity especially in rainfed areas. (60% of the country's net sown area is rainfed and accounts for 40% of the total food production.)

National Mission for Sustainable Agriculture

- Stated dimensions of NMSA:
    a. Improved crop seeds, livestock and fish cultures
    b. Water Use Efficiency
    c. Pest Management
    d. Improved Farm Practices
    e. Nutrient Management
    f. Agricultural insurance
    g. Credit support
    h. Markets
    i. Access to Information
    j. Livelihood diversification

Other Initiatives

Soil Health Card (SHC) Scheme

- SHC scheme is under implementation since 2015 to provide Soil Health Card to all farmers every two years. It will provide information to farmers on soil nutrients status of their soil and recommendation on appropriate dosage of nutrients to be applied for improving soil health. •

- It contains the status of soil with respect to 12 parameters:
  - N-P-K **(Macro-nutrients)**;
  - Sulphur **(Secondary-nutrient)**;
  - Zn, Fe, Cu, Mn, Bo **(Micro-nutrients)**; and
  - pH, EC, OC **(Physical parameters)**.

- Based on this, the SHC will indicate fertilizer recommendations and soil amendment required for the farm.

Other Initiatives

National Mission on Natural Farming

- National Mission on Natural Farming or Bharatiya Prakritik Krishi Paddhati Programme (BPKP) and is a sub-mission under centrally sponsored scheme Paramparagat Krishi Vikas Yojana (PKVY).

- PKVY falls within the umbrella of the **National Mission on Sustainable Agriculture (NMSA).**

- BPKP is being upscaled as '**National Mission on Natural Farming (NMNF)**/ (Bhartiya Prakratik Krishi Paddhati)' for implementation all across the country.

- **Tenure:** 6 years (2019-20 to 2024-25).

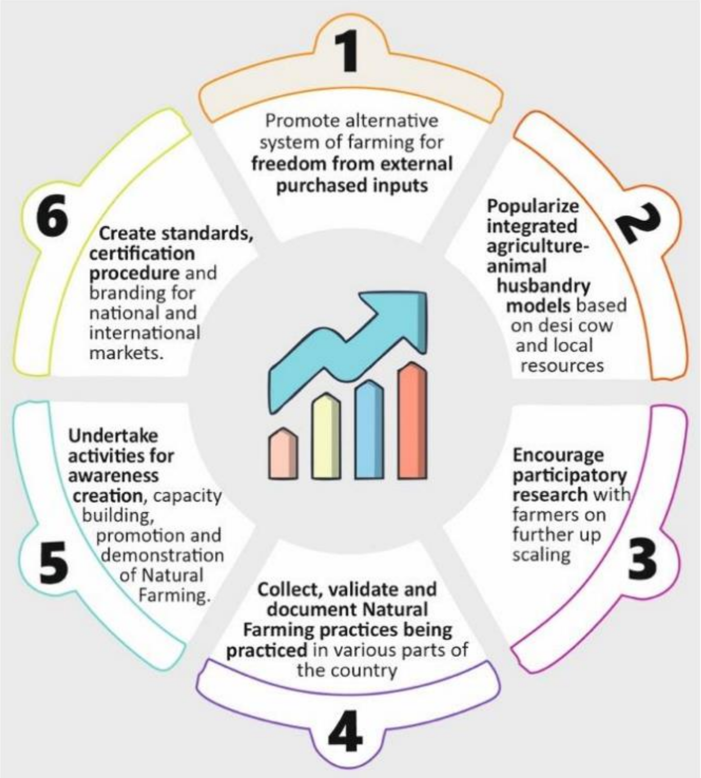
Other Initiatives

National Mission on Natural Farming

- It will be a demand driven programme and states shall prepare a long-term perspective plan with year-wise targets and goals.
- **Knowledge partner for natural farming extension:** National Institute of Agricultural Extension Management (MANAGE).
- **National Centre of Organic and Natural Farming (NCONF)** shall work towards development of certification programme for Natural Farming, establish secretariat for certification management, portal development, management, maintenance and integration with other portals.

Other Initiatives

Objective of the Mission



Other Initiatives

COMPONENTS OF NATURAL FARMING

Beejamrit

The process includes **treatment of seed** using cow dung, urine and lime based formulations.

Contact

The process involves **activating earthworms in the soil** in order to create water vapor condensation.

Jivamrit

The process enhances the fertility of soil using cow urine, dung, flour of pulses and jaggery concotion.

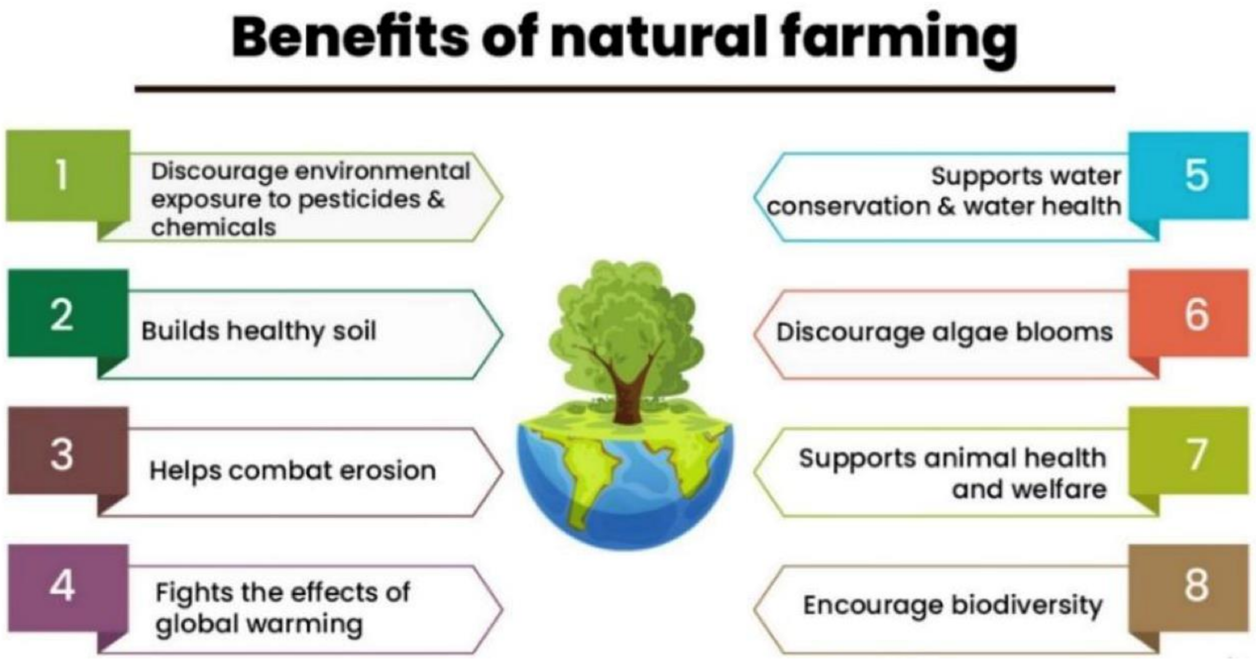
Mulching

The process involves **creating micro climate using different mulches** with trees, crop biomass to conserve soil moisture.

Plant Protection

The process involves **spraying of biological concoctions** which prevents pest, disease and weed problems and protects the plant and improves their soil fertility.

Other Initiatives



Other Initiatives

SIMILARITY BETWEEN ORGANIC FARMING AND NATURAL FARMING

- Both are non-chemical system of farming.
- Both are based on diveristy, on-farm biomass management and biological nutrient recyling.
- Diversity, rotation multiple cropping and resource recycling is key.

Differences between Organic Farming and Natural Farming

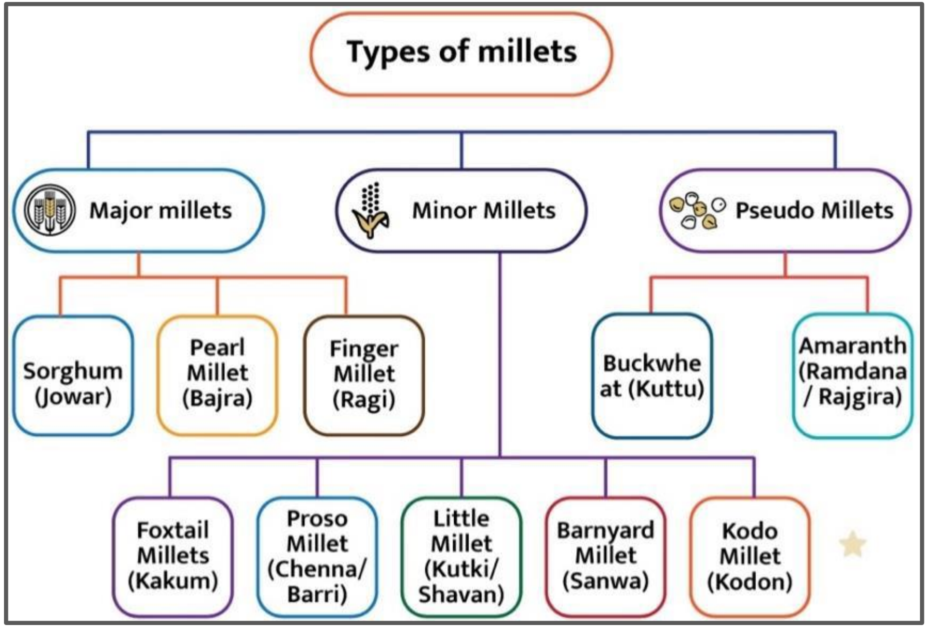
Organic Farming	Natural Farming
Open for use of off-farm organic and biological inputs	No external inputs, only on-farm inputs based on Desi Cow (Jeevamrit, Beejamrit, Ghanajeevamrit)
Open for micronutrient correction through use of minerals	Use of compost/vermicompost and minerals are not allowed
Plowing, tilling, mixing manure, weeding, and other fundamental agro activities are still required.	There is no plowing, no soil tilling, no fertilizers, and no weeding.
Widely popular, Global market at 132 billion US$	Evolving markets are yet to be developed

Other Initiatives

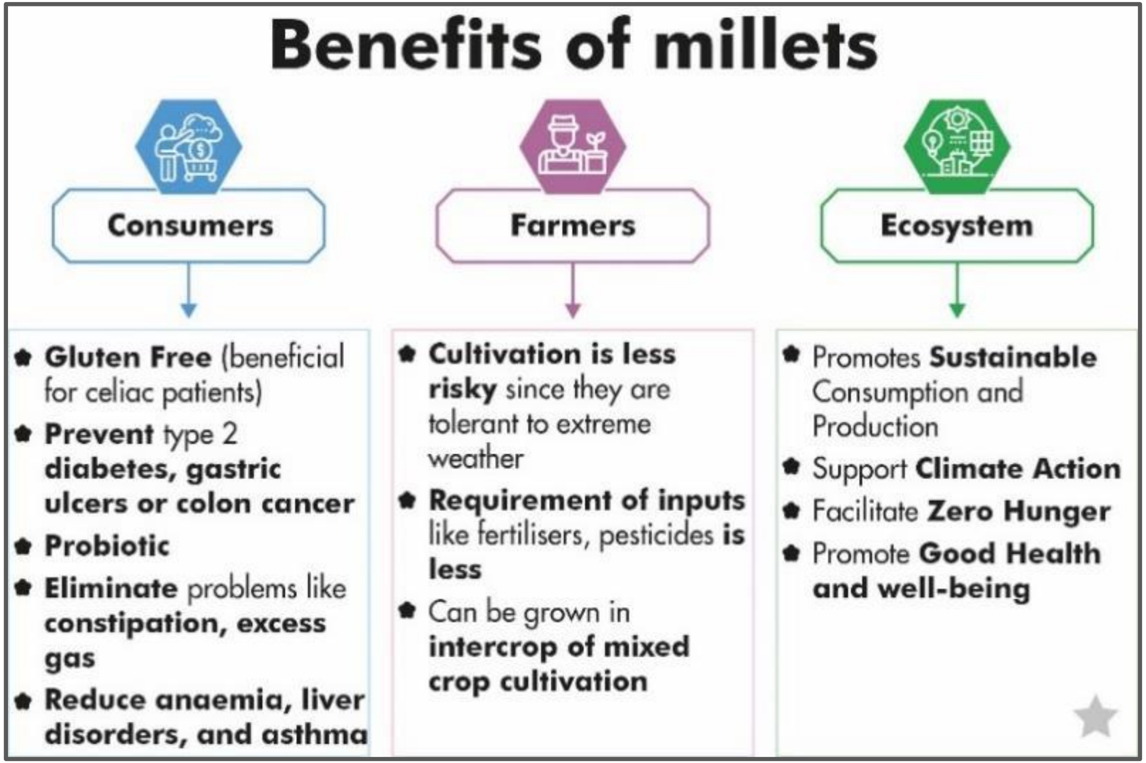
Millets

- A **diverse family** of small-grained cereals(Poaceae family), indigenous to various parts of India.
- Popularly known as **Nutri-cereals** as they provide most of the nutrients required for normal functioning of the human body.
- Contain 7-12% protein, 2-5% fat, 65-75% carbohydrates and 15-20% dietary fibre.
- Before the **Green Revolution**, millets were one of the **largest grown staples** in India, but have been reduced to a marginal fodder crop to feed livestock.
    - India produces **20% of the globe's production** that is led by Africa and the Americas.
    - India exports millets products worth of USD 34.32 million during 2021- 22.

Other Initiatives



Other Initiatives



Other Initiatives

- Top 5 millet-producing states in India- **Madhya Pradesh**, Gujarat, Karnataka, Rajasthan, and Maharashtra.

Initiatives to promote millet

- **National year of millets** was observed in **2018**
- Increase in Minimum Support Price (MSP) to support millet cultivators.
- Saksham Anganwadi and **Poshan 2.0** mandates supply of millets at least once a week in midday meal scheme.
- Recently, the Food and Agriculture Organization (FAO) of the United Nations, organized an opening ceremony for the International Year of Millets – 2023 (IYM2023) in Rome, Italy.

Other Initiatives

- **International Year of Millets (IYM) 2023**
- The United Nations 2021 has adopted **India's proposal** to declare **2023** as the IYM.
- The IYM 2023 will **raise awareness** about the importance of millets in food security and nutrition.
- The IYM 2023 will provide an opportunity to:
    - increase **global production,**
    - efficient processing and better use of crop rotation,
    - promote millets as a major component of the food basket,
    - promote research and development on millet.

Other Initiatives

Direct-seeded rice (DSR)

- It refers to the process of establishing a rice crop from seeds sown in the field rather than by transplanting seedlings from the nursery.
- It is a water-saving method of sowing paddy.
- It is also known as broadcasting seed technique.
- has been rejected by Farmers in Punjab.

Other Initiatives

Advantages of DSR	Issues with DSR
• **Saves labour.** • Sowing can be done in stipulated time frame because of easier and faster planting. • **Early crop maturity by 7-10 days** which allows timely planting of subsequent crops. • **More efficient water use** and higher water stress tolerance. • Less methane emission.	• **Uses more seed** than transplanting. • **Laser land levelling** costs Rs 1,000/acre is compulsory in DSR. This is not so in transplanting. • Some direct-sown crops may be harder to get started in cold (or hot) conditions. • Non- availability of the herbicides. • **Seed requirement for DSR is higher, at 8-10 kg/acre,** compared to 4-5 kg in transplanting.

Other Initiatives

OTHER METHODS OF RICE CULTIVATION

System of Rice Intensification (SRI)
First developed in Madagascar, save 15 to 20% ground water, improves rice productivity.

Transplantation Seeds
First sown in nursery and seedlings are transplanted to main field once they show 3-4 leaves. It requires heavy labour

Drilling method
One person ploughs a hole in land and other person sows the seed. Or is most commonly used to plough the land.

Japanese method
Seeds are sown in nursery beds and then transplanted to the main field.

National Mission for A Green India

- The Green India Mission aims to sequester 2.523 billion tonnes of carbon by 2020-30, and this involves adding 30 million hectares in addition to existing forest.
- It aims at protecting; restoring and enhancing India's diminishing forest cover and responding to climate change by a combination of adaptation and mitigation measures.
- The mission will be implemented on both public as well as private lands. It will engage local communities in planning, decision making, implementation and monitoring.

National Mission for A Green India

- The intended major outcomes of the project :
  - Improved ecosystem services Reversal of land degradation.
  - Improvement in quality of forest cover and ecosystem services of forests, degraded grassland and wetlands.
  - Eco-restoration of shifting cultivation areas, cold deserts, mangroves, ravines and abandoned mining areas.
  - Improvement in forest and tree cover in urban lands.
  - Improvement in tree cover on agricultural lands and other non-forest lands (agroforestry/social forestry).

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National Mission on Seabuckthorn

- The MoEF and DRDO have launched the initiative for Seabuckthorn cultivation in the cold deserts. The initiative is a part of **Sub-Mission on Cold Desert Ecosystems** under the Green India Mission.

Seabuckthorn

- Seabuckthorn, popularly known as Leh berries is also called the "Wonder plant" and "Ladakh gold".
- It has multi-purpose medicinal and nutritional properties.
- The plant has the ability to fix atmospheric nitrogen.
- It is tolerant to extreme temperatures and has an extensive root system, making it ideal for controlling soil erosion and preventing desertification.

Other Initiatives

“Cloud Forest Assets Financing a Valuable Nature-Based Solution"

- Recently a new report **“Cloud Forest Assets Financing a Valuable Nature-Based Solution"** was released by **Earth Security,** a global nature-based asset management advisory firm
- The suggested Cloud **forest bonds** as per the report are a part of ‘ Nature Based Solutions (NBS)’ and their financing to protect these Cloud forests.

Cloud Forests

- Cloud Forests are mountain tropical forests generally found at **the river headstreams and mostly covered with clouds.**
- These forests serve as the storage of clean water for communities, industries and hydropower plants.
- Majority of Cloud Forest i.e., 90% are found in 25 developing countries in tropical regions which bears the disproportionate impact of climate change.

Other Initiatives

What is a cloud forest?

Areas with forest cover greater than 10% and fog present greater than 70% of the time.

Found mostly at between 1,500 and 3,000 above sea level.	Shorter trees covered in mosses and ferns.
Cooler and wetter than other forests.	Blanketed in dense ground-level clouds.

Other Initiatives

Benefits of Cloud forest bonds

- Resource Monetization
- Prospecting true potential of ecosystem services
- Stepping Towards Green Economy
- Combating Climate Change
- Alternate Climate Financing Mechanism for low income countries
- Participation and Community Capacity Building

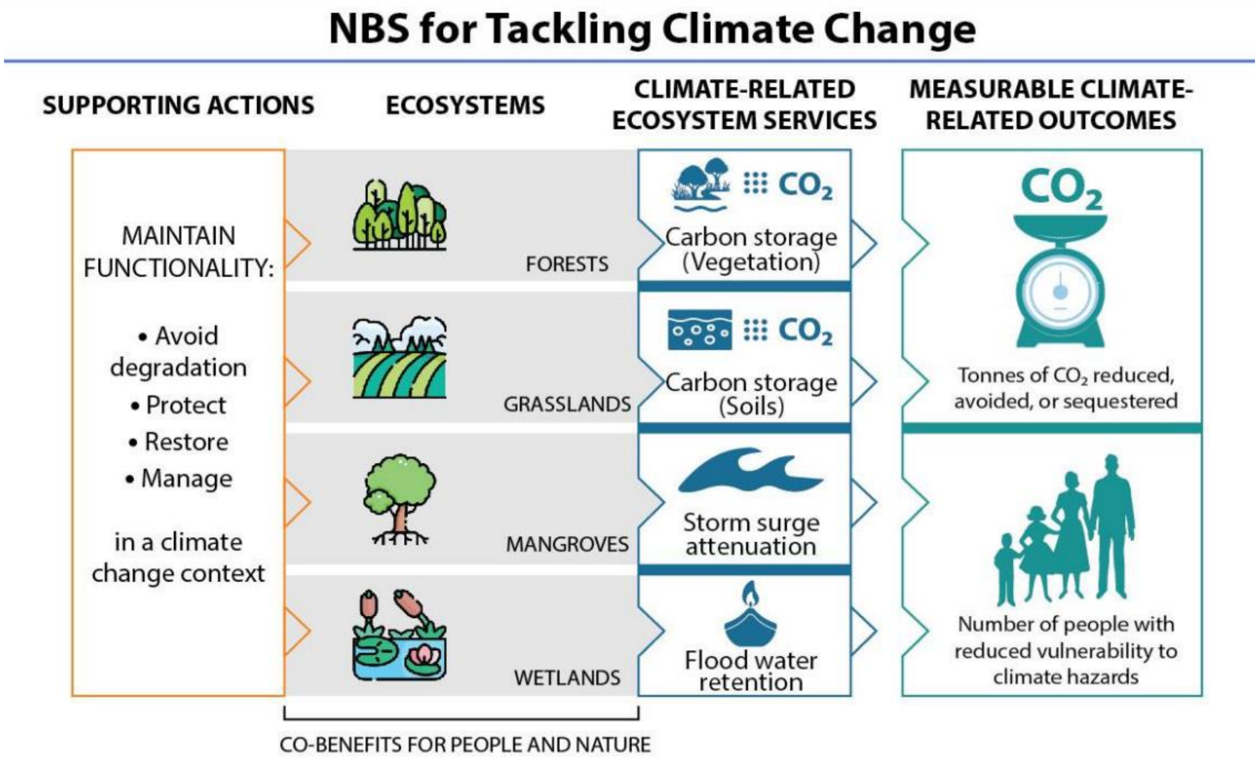
Cloud Forest and The Cloud Forest 25 (CF25) initiative

- CF25 is an Investment Initiative to bring countries, their creditors and multilaterals organizations together to accelerate, and consolidate the progress and scale such investment products.
- **Financing Mechanisms:**
  - **Water Payments:** To create a mechanism for payments for ecosystem services from water users such as hydropower dams which works on a national scale and is subject to compliance norms.
  - **Sovereign Carbon:** The financing of forest carbon at sovereign and sub-sovereign jurisdictional scales as part of an approach to wider areas of lowland tropical rainforests.
  - **Cloud forest bonds:** They are debt based instruments to mobilise finance for protection of cloud forest. Their value is based on the economic value of a country's Cloud forest resources.

Nature-based solutions

- Nature-based solutions are ways of addressing environmental challenges and achieving sustainable development by working with, and enhancing, nature.
- These solutions use the power of ecosystems and natural resources to provide benefits for both people and the environment.
- Examples of nature-based solutions include reforestation, ecosystem restoration, sustainable agriculture, green infrastructure, and natural flood control measures.
- Nature-based solutions can help address a wide range of challenges, such as climate change, biodiversity loss, water scarcity, food security, and natural disasters.
- They often provide multiple benefits, such as improving soil health, reducing greenhouse gas emissions, and enhancing biodiversity.

Other Initiatives



National Mission on Strategic Knowledge for Climate Change (NMSKCC)

- It seeks to build a knowledge system that supports national policy and action for responding effectively to climate change challenges, while not compromising on the nation's growth goals.

- **Mission Objectives:**
  - Formation of knowledge networks engaged in research and development relating to climate science.
  - Development of national capacity for modelling the regional impact of climate change.
  - Establishing research networks and encouraging research in the areas of climate change impacts on important socio-economic sectors like agriculture, health, ecosystems, biodiversity, coastal zones, etc

India’s Initiatives

Mission Innovation

- Global initiative to catalyze action and investment in research, development and demonstration to make clean energy affordable, attractive and accessible to all this decade.
- It consists of 23 countries and the European Commission (on behalf of the European Union)
- **India** is a **founding member.**
- First phase of the mission launched alongside the Paris Agreement in 2015.
- Mission Innovation 2.0, second phase of MI, was launched in 2021.
- MI has an Innovation Platform for member countries to track innovation progress, exchange knowledge and work with investors, innovators and end-users to accelerate technologies to market

India's Initiatives

India's Initiatives

Mission Integrated Bio-refineries (MIB)

- 7th mission under Mission Innovation (MI). Other 5 missions includes: **Clean Hydrogen, Green Powered Future, Zero-Emission Shipping, Carbon Dioxide Removal and Urban Transitions.**
- Launched in **April 2022.**
- Aim: Greater international collaboration and financing for Energy RD&D during the next five years.
- **Members:**
  - **Co-lead: India** and **Netherlands**
  - Core mission members: Brazil and Canada
  - Mission support group: European Commission and UK.

India’s Initiatives

- **Goal:** To develop and demonstrate innovative solutions to accelerate the commercialization of integrated biorefineries.
- **Target:** To replace **10% of fossil-based fuels**, chemicals and materials with **bio-alternatives** by 2030
    - [Example of bio-alternative- oil extracted from the seeds of the plant, **Jatropha** is now being used as a bio alternative to diesel]
- Prioritizes eight collaborative actions organized around 3 Pillars of
    - Supporting Research, Development and Demonstration (RD&D).
    - Accelerating Pilots and Demonstrations.
    - Improving Policy and Market Conditions.

India’s Initiatives

Innovation Roadmap of the Mission Integrated Biorefineries

- **Context:** India announced launch of ‘Innovation Roadmap of the Mission Integrated Biorefineries’ (IRMIB) at Global Clean Energy Action Forum.
    - **Global Clean Energy Action Forum**, is a joint convening of the 13th Clean Energy Ministerial and the 7th Mission Innovation ministerial.
- **Developed by:** Co-leads of Mission Integrated Biorefineries (MIB) with active inputs from Brazil, Canada, EC and the UK.
- Aims to fill the void by:
    - Identifying gaps and challenges in current biorefining value chains,
    - Prioritising Eight key actions to support the Mission,
    - Guiding the Mission's overall path in achieving its goal.

India's Initiatives

CleanTech Exchange

- **News-** Recently, India launched Mission Innovation (MI) - CleanTech Exchange under the Innovation Platform of Mission Innovation.
- It was launched virtually at the Innovating to Net Zero Summit hosted by Chile last year.
- **About CleanTech Exchange**
    - CleanTech Exchange is a global initiative to create a **network of incubators across member countries** to accelerate clean energy innovation.
    - The network will provide access to the expertise and market insights needed to support new technologies to access new markets globally.

India's Initiatives

Mission LiFE (Lifestyle for Environment)

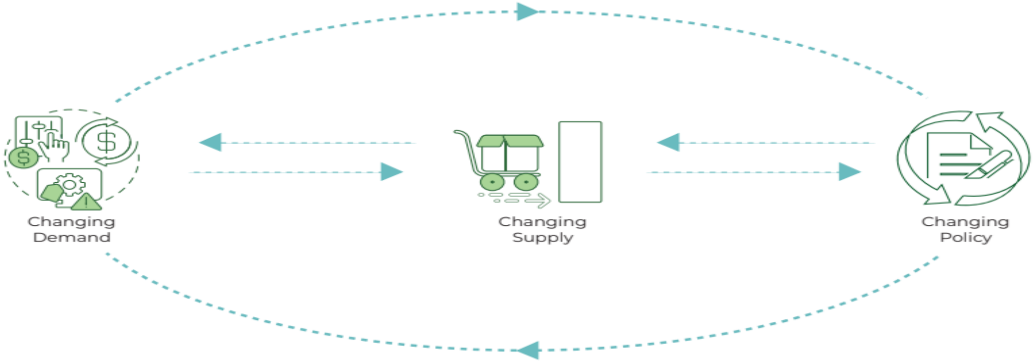
- **Context-** The Prime Minister (PM) on 20th October 2022 launched Mission LiFE (Lifestyle for Environment) at the Statue of Unity at Kevadia, Gujarat to protect the environment.
- It is an **India-led** global mass movement.
- Aim: To nudge individual and collective action to protect and preserve the environment.
- Concept introduced by the Indian PM at the **UNFCCC COP-26 in Glasgow**.
- **Targets:** Mobilise at least **1 billion Indians** and other global citizens to take individual and collective action for protecting and preserving the environment in the period 2022 to 2027.

India’s Initiatives

- Within India, at least 80% of all villages and urban local bodies aimed to become environment friendly by 2028.
- **Mission Duration:** **5-year programme** visualizing three core shifts in our collective approach towards sustainability.
    - In 2022-23, the mission will focus on Phase I.
- **Implementation:** **NITI Aayog** to curate and incubate in the first year.
- Subsequently implementation by the Ministry of Environment, Forest and Climate Change (MoEFCC) in a non-linear and nonsequential manner.
- **Vision:** To live a lifestyle that is in tune with our planet and does not harm it. People living such a lifestyle are called “Pro-Planet People (P3)”
- Lists **75 lifestyle practices** under **7 categories**

India’s Initiatives

- **Change in Demand (Phase I):** Nudging individuals across the world to practice simple yet effective environment-friendly actions in their daily lives.
- **Change in Supply (Phase II):** Changes in large-scale individual demand are expected to gradually nudge industries and markets to respond and tailor supply and procurement as per the revised demands.
- **Change in Policy (Phase III):** By influencing the demand and supply dynamics of India and the world, the long-term vision of Mission LiFE is to trigger shifts in large-scale industrial and government policies that can support both sustainable consumption and production.



India’s Initiatives

India’s Initiatives

Impact

When estimated against a business-as-usual scenario by 1 billion Indians in 2022-23 to 2027-28, the impact of LiFE actions can be significant, as demonstrated below with select examples:

- Switching off the car / scooter engines at traffic lights / railway crossings can save up to **22.5 billion kWh of energy**
- Turning off running taps when not in active use can save upto **9 trillion litres of water**
- Using a cloth bag instead of a plastic bag while shopping can save up to **375 million tonnes of solid waste** from entering the landfill
- Discarding non-functioning gadgets in the nearest e-recycling unit can recycle upto **0.75 million tonnes of e-waste**
- Composting waste food at home can save up to **15 billion tonnes of food** from going to landfills

India’s Initiatives

Forum for Decarbonizing Transport

- **Context-** NITI Aayog and World Resources Institute (WRI) India Jointly Launch ‘Forum for Decarbonizing Transport’ in India.
- It is part of the **Nationally Determined Contribution-Transport Initiative for Asia (NDC-TIA) project.**
- NDC-TIA is a **joint programme** of **seven organisations** that will engage China, India, and Vietnam with the objective to facilitate a paradigm shift to zero-emission transport across Asia.
- The project is part of the **International Climate Initiative (IKI)** and **NITI Aayog** is the implementing partner for India.
- Transport in India is the third most CO_2 emitting sector.

Clean Energy Ministerial's (CEM) – Industrial Deep Decarbonization Initiative (IDDI)

- **Launched by:** India and UK
- IDDI is a global coalition of public and private organisations who are working to stimulate demand for low carbon industrial materials.
- It is coordinated by UNIDO and countries like Germany and Canada have also joined the initiative.
- It works to
  - Standardise carbon assessments
  - Establish ambitious public and private sector procurement targets
  - Incentivise investment into low-carbon product development and design industry guidelines

India’s Initiatives

- It brings together a strong coalition of related initiatives and organizations to tackle carbon intensive construction materials such as steel and cement like
    - The Mission Possible Platform
    - The Leadership Group for the Industry Transition
    - The International Renewable Energy Agency (IRENA)
    - World Bank

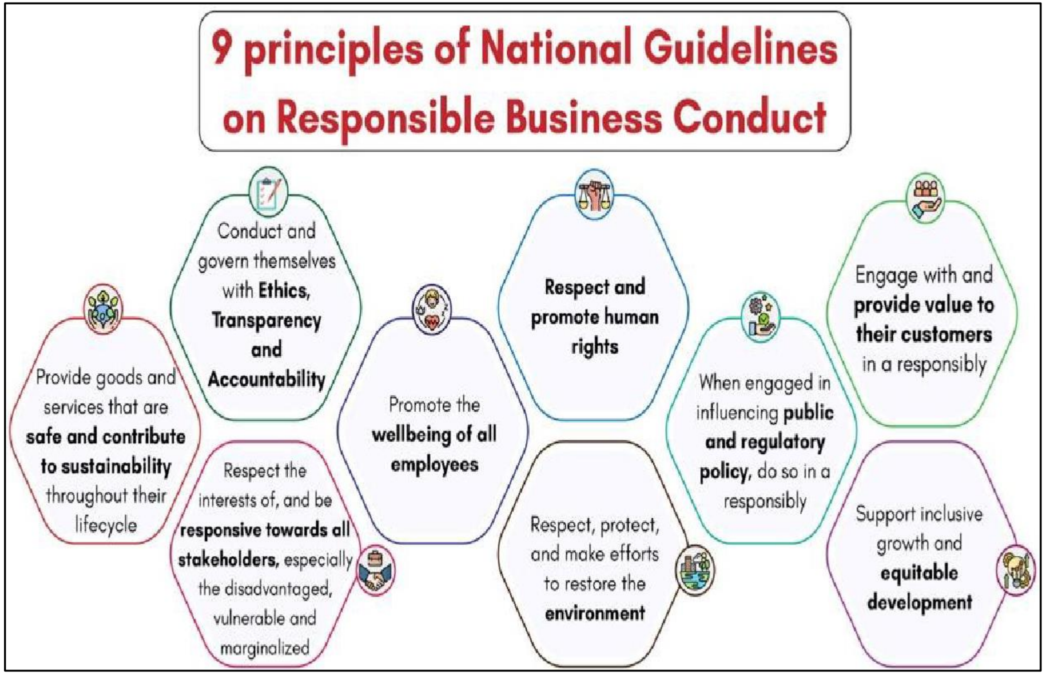
India’s Initiatives

Environmental, Social and Governance (ESG)

- ESG is a framework that helps stakeholders understand how an organization is managing risks and opportunities related to environmental, social, and governance criteria.
- Environmental criteria consider how a company performs as a steward of nature.
- Social criteria examine how it manages relationships with employees, suppliers, customers, communities where it operates etc.
- Governance deals with a company’s leadership, audits, internal controls, shareholder rights etc.
- In 2021, SEBI issued new sustainability reporting requirements under **Business Responsibility and Sustainability Report (BRSR).**

India's Initiatives

- BRSR aims to establish links between financial results of a business with its ESG performance.
- BRSR was made mandatory for **top 1,000 listed entities** (by market capitalisation) from 2022–23.
- BRSR seeks disclosure from listed entities on their performance against nine principles of 'National Guidelines on Responsible Business Conduct' (NGBRCs).



India’s Initiatives

In news

- Recently Securities and Exchange Board of India (SEBI) proposed Framework on ESG (Environmental, Social and Governance) Disclosures, Ratings and Investing.
- Proposed ESG disclosures by listed entities, ESG Ratings in securities market and ESG Investing by Mutual Funds will facilitate balance between transparency, simplification and ease of doing business in ESG domain.

India’s Initiatives

- Key features of the disclosure

ESG Disclosures	• Updating Business Responsibility and Sustainability Reporting (BRSR) to **incorporate missing Key Performance Indicators KPIs** in BRSR core. ○ BRSR establish links between financial results of a business with its ESG performance. • **Mandating reasonable assurance** of **KPIs** in **BRSR core** through **glide path approach.** • Introduce a limited set of ESG disclosures for **supply chain.** 
ESG Ratings	• **Inclusion** of **15 ESG parameters** with an Indian context by **ESG Rating Providers (ERPs)** in ratings. ○ This includes Indian standards like Extended Producer Responsibility (EPR) scheme, more comprehensive gender diversity, etc. • ERPs also provide **Core ESG rating** based on information/reports that are assured/audited/verified.
ESG Investing	• **Asset Management Companies (AMCs)/Mutual Funds** should provide better clarity on 'in favor' or 'against' votes cast on resolutions due to any ESG reason. • To mitigate mis-selling and **Greenwashing**, an **ESG scheme** should invest **at least 65%** of its **asset under management (AUM)** in companies reporting on comprehensive BRSR and providing assurance on BRSR Core disclosures.

India’s Initiatives

India’s First Sovereign Green Bonds (SGB) Framework.

- **Context:** Ministry of Finance approves India’s First Sovereign Green Bonds (SGB) Framework.

About Sovereign Green Bonds

- A green bond is a fixed-income instrument designed to support specific climate-related or environmental projects.
- SGBs are issued by the Government.
- Earlier, Union Budget 2022-23 announced the issuance of SGBs.

India’s Initiatives

Key highlights of framework:

- A ‘green project’ classification is based on the principles like: Encourages energy efficiency, reduces carbon & GHG emissions etc.
- Eligible projects under the framework include: Renewable energy, clean transportation, water and waste management, green building etc.
- **Excludes,** nuclear power generation, landfill projects, direct waste incineration, hydropower plants larger than 25 MW etc.
- Green Finance Working Committee constituted to validate key decisions on issuance of SGB.
- Proceeds will be deposited with **Consolidated Fund of India.**

India's Initiatives

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Municipal Green Bonds

- SEBI has announced that **issuers of municipal debt securities** can issue green bonds in compliance with guidelines for issuance and listing non-convertible debentures.
- Eligible projects include, renewable energy, clean transportation, water and waste management, green building etc.

Blue Bonds

- SEBI has proposed the concept of blue bonds as a mode of sustainable finance.
- A blue bond is a relatively new form of debt instrument that is issued to support investments in healthy oceans and blue economies.
- Blue Economy is sustainable use of ocean resources for economic growth, improved livelihoods, and jobs.

India’s Initiatives

- Blue bonds offer an opportunity for mobilising private sector capital to be mobilized to support the blue economy.
- India can deploy blue bonds in various aspects of blue economy like oceanic resource mining, sustainable fishing, national offshore wind energy policy etc.
- The Republic of Seychelles had launched the world’s first sovereign blue bond in 2018.
    - **Yellow bonds:** Funds raised for solar energy generation and upstream industries and downstream industries associated with it.
    - **Transition bonds:** Funds raised for transitioning to a more sustainable form of operations, in line with India’s Intended Nationally Determined Contributions.

SUSTAINABLE DEVELOPMENT REPORT 2022

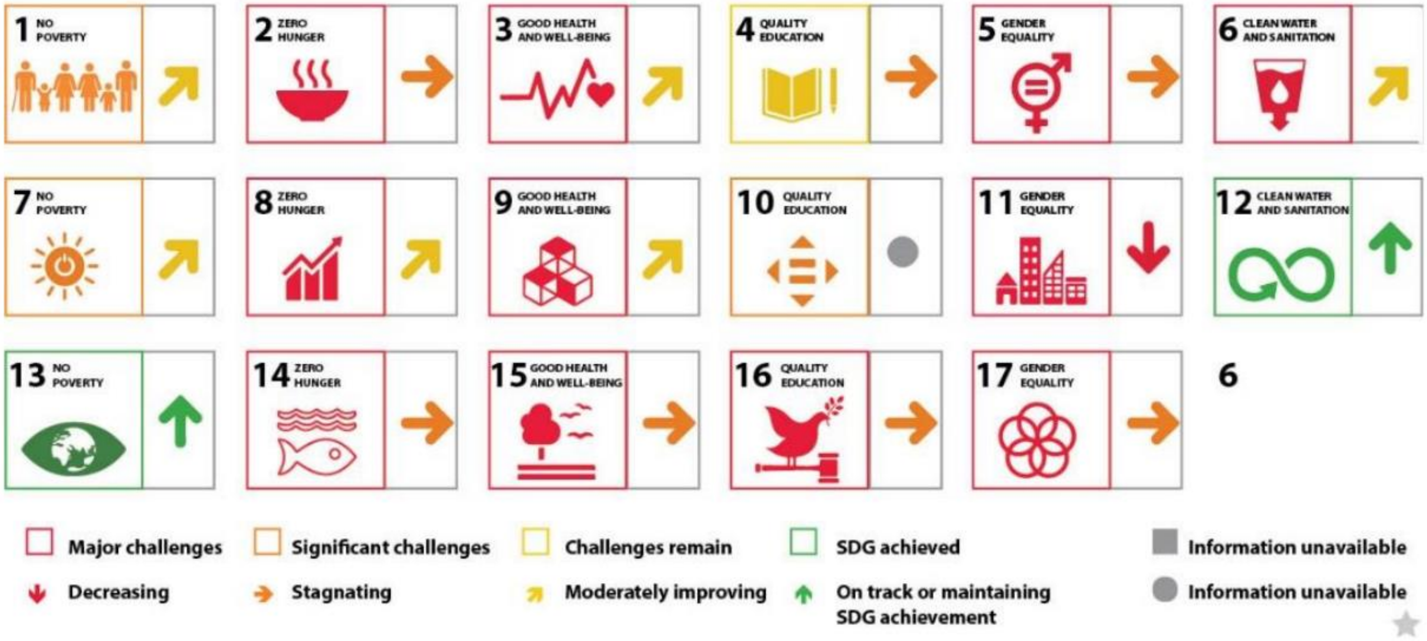
- India ranked at 121 out of 163 countries in the Sustainable Development Report (SDR), 2022, published by a group of independent experts at the **Sustainable Development Solutions Network (SDSN).**

India's Performance

- India's rank slipped for the 3rd consecutive year from 117 in 2020 and 120 in 2021.
- India achieved 2 SDG goals (SDG 12 & 13), challenges remain in 1(SDG 4), significant challenges remain in 3 (SDG 1, 7 & 10) and major challenges remain in 11 out of total 17 SDGs (SDG 2, 3, 5,6,8,9,11,14,15,16 & 17).

Sustainable Development

India's Performance on SDGs



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India-UN Sustainable Development Cooperation Framework 2023–27

- The framework is the single most important instrument capturing the entirety of **26 UN entities' plans and programmes in India.**
- It aims to align the four pillars of the 2030 Agenda—People, Prosperity, Planet and Participation, with India's national priorities.
- It has six outcome areas:(i) health and well-being (ii) nutrition and food (iii) quality education (iv) economic growth and decent work (v) environment, climate, wash and resilience (vi) empowering people, communities, and institutions.
- It takes up from the GoI–UNSDF 2018–22 and will be signed as India completes 75 years of independence.

SUSTAINABLE DEVELOPMENT REPORT 2022

Key Findings of the report:

- For the second year in a row, **world is no longer making progress on the SDGs.**
- Based on **2022 International Spillover Index,** rich countries generate negative international spillovers notably through unsustainable consumption.
    - **Spillover effect refers to effect on the economy of a country from unrelated events happening in another country.**
- Peace, diplomacy, and international cooperation are fundamental conditions for the world to progress on the SDGs towards 2030 and beyond.

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Gross Environment Product (GEP)

- **<u>Context</u>**
  - **<u>Uttarakhand</u>** became the first state in India to take into account Gross Environment Product (GEP) while calculating its Gross Domestic Product (GDP).
  - Four critical natural resources- **<u>Air, Water, Forest and Soil</u>**- will be assigned monetary values.
  - The quality and quantity of these natural resources would determine the GEP of Uttarakhand

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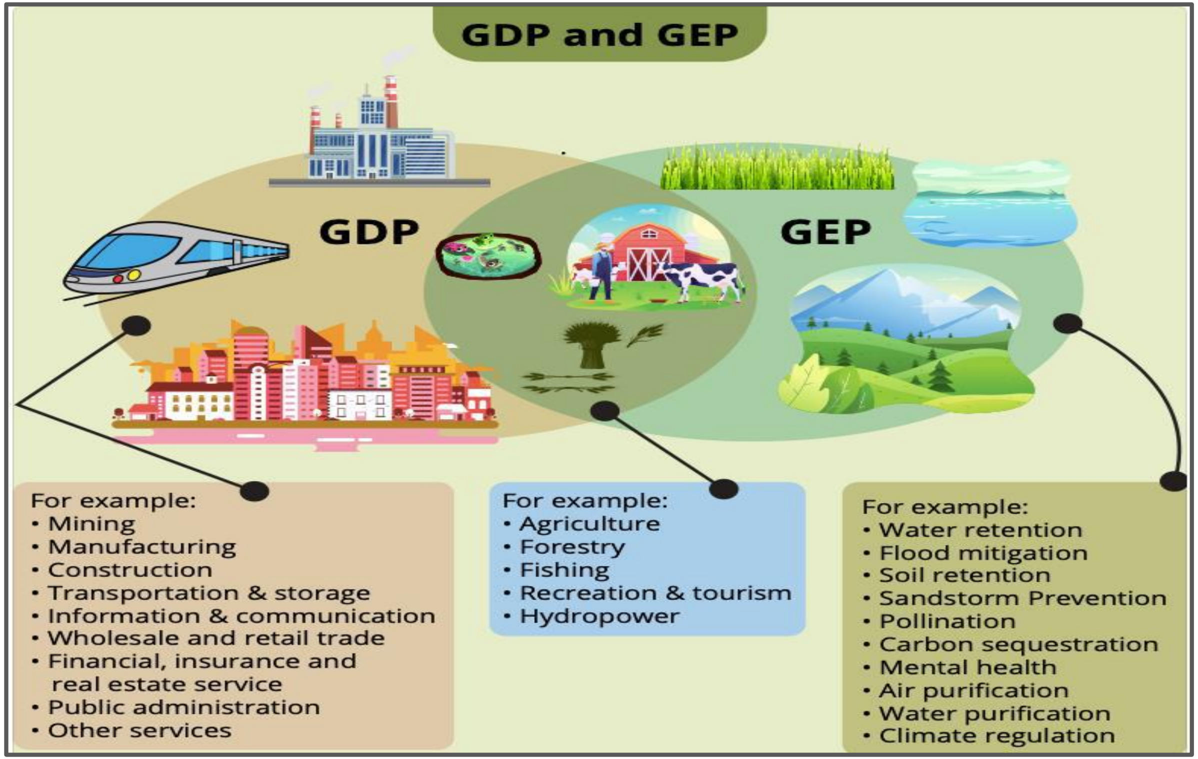
About GEP:

- It is the **total value of final ecosystem services** supplied to **human well-being** in a region annually and can be measured in terms of biophysical value and monetary value.
- It indicates the **overall health** of the environment as GEP measures prime indicators such as forest cover, soil erosion, air quality and dissolved oxygen in river water.
- Unlike **Green GDP** which is obtained after **deducting the damage** to the environment from the total production of the state, GEP will assess the **improvement** in the environment components in a year.
- Further it will tell how much work the state has done in reducing the loss of the ecosystem in environmental protection and resource use.

Sustainable Development



Sustainable Development

Other global standards /initiatives	
System of Environmental and Economic Accounts (SEEA)	Guidebook developed by the United Nations to provide standards for incorporating natural capital and environmental quality into national accounting systems.
Happy Planet Index (HPI)	It was created by the **British New Economics Foundation (NEF) to** measure national welfare in the context of environmental sustainability.
Bhutan's Gross National Happiness (GNH)	It has environmental preservation as one of the four policy objectives.
Other	• **China** (since 2004) **has been undertaking studies to estimate the cost of various types of environmental damage which offsets its economic growth.** • **Sweden** (since 2003) **has brought in various environmental indicators** (like air emissions, waste etc.) as part of the government policy of achieving sustainable development

Circular Economy

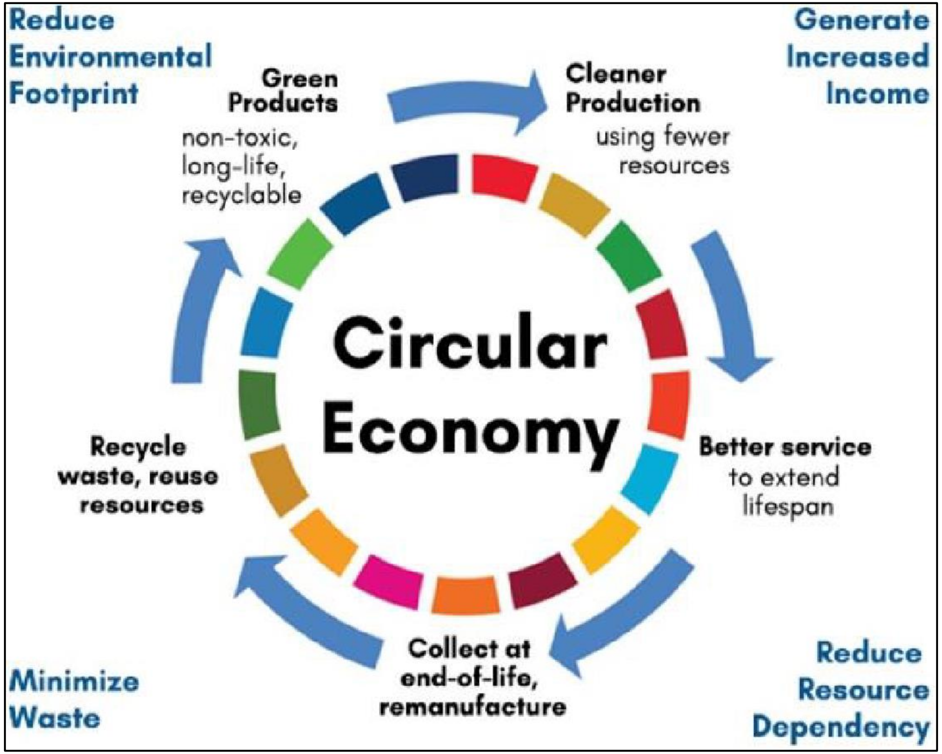
- **Context-** United Nations Development Programme (UNDP) India launched a campaign to drive inclusive circular economy.

A **linear economic model of production and consumption,** natural resources are turned into products that are ultimately destined to become waste because of the way they have been designed and manufactured. This process is summarised by **"take, make, waste".**	A **circular economy** employs reusing, sharing, leasing, repairing, refurbishing, remanufacturing and recycling to create a **closed-loop system,** minimising the use of resource inputs and the creation of waste, pollution and carbon emissions

Sustainable Development

Circular Economy

- It involves sharing, leasing, reusing, repairing, refurbishing and recycling existing materials and products as long as possible to:
- Extend life cycle of products.
- Reduce waste to a minimum by creating further value.



Sustainable Development

UNDP Initiative will focus on:

- End-to-end management of plastic waste by promoting segregation of waste at source and collection of segregated waste.
- Setting up **Material Recovery Facilities (MRFs)** or Swachhata Kendras for recycling all kinds of plastic waste along the value chain.
- Social Inclusion of 20,000 Safai Saathis or waste pickers through access to government welfare schemes and linkages etc.

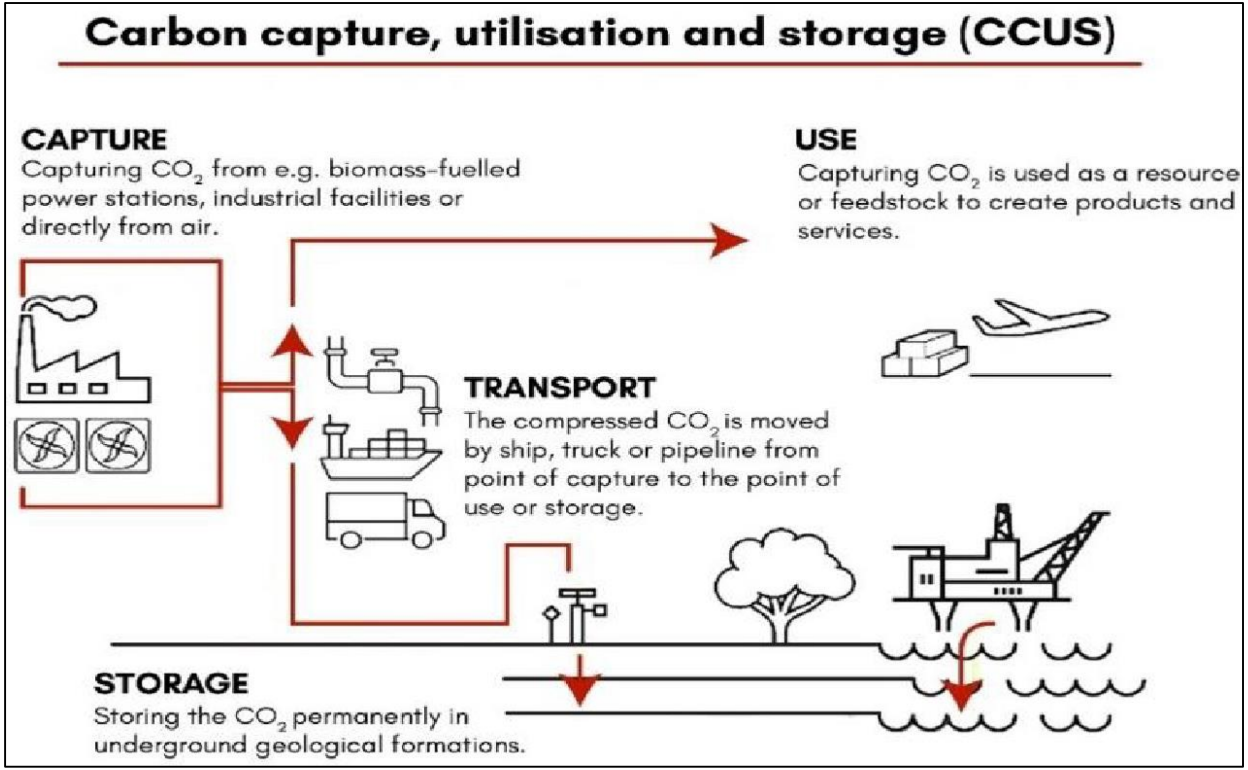
Sustainable Development

- Building capacities of Urban Local Bodies for adopting MRFs model for plastic and dry waste management
- Initiative is a scale-up of existing partnership under **UNDP's flagship Plastic Waste Management Programme** to develop a sustainable model for plastic waste management in India.
- It promotes collection, segregation, and recycling of all plastics to move towards a circular economy

Carbon capture utilisation and storage (CCUS)

- Encompasses technologies to remove CO_2 from flue gas and the atmosphere, followed by recycling the CO_2 for utilization and determining safe and permanent storage options.
  - [**Flue gases** are produced when coal, oil, natural gas, wood or any other fuel is combusted in an industrial furnace, a power plant's steam-generating boiler, or other large combustion device.]
- CO_2 captured using CCU technologies are converted into fuel (methane and methanol), refrigerants, building materials, **blue hydrogen** etc.
- Captured gas is used directly in fire extinguishers, pharma, food and beverage industries as well as the agricultural sector.

Sustainable Development



- **Different Carbon Capture technologies for different applications are as follows:**

Technology	Details
Chemical Solvent	• Preferred when dealing with gas streams that are lean in CO_2. • Have relatively lower pressures such as flue gas streams from power plants etc.
Physical Solvent	• Work well on gas streams with relatively higher CO_2 concentration and pressure such as pre-combustion capture in case of gasification projects
Adsorption	• Suitable for gas streams with moderate to high pressure and moderate CO_2 concentration such as steam methane reforming (SMR) flue gas.
Cryogenic Separation	• Preferred in cases where cost of power is low.

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Initiative for CCUS in India

- **National Centres of Excellence:** Establishment of two National Centres of Excellence in CCUS at IIT Bombay and JNCASR, Bengaluru.
    - These centres will facilitate capturing & mapping of current R&D and innovation activities in the domain and also develop networks of researchers, industries and stakeholders.
- **Mission Innovation Challenge** on CCUS to enable near-zero CO2 emissions from power plants and carbon-intensive industries.
- **Accelerating CCS Technologies (ACT)** to facilitate R&D and innovation that can lead to development of safe and cost-effective CCUS technologies.

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Recent news

- **NITI Aayog** launched carbon capture utilisation and storage (CCUS) policy framework and its deployment mechanism in India.
- **About the policy framework**
    - Aims to develop and implement a practicable framework to accelerate research and development on CCUS in India.
    - Explores the importance of the technology as an emission reduction strategy to achieve deep decarbonization from the hard-to-abate sectors like steel, cement etc.
    - More discussion on CCUS to follow in **National Hydrogen Mission**

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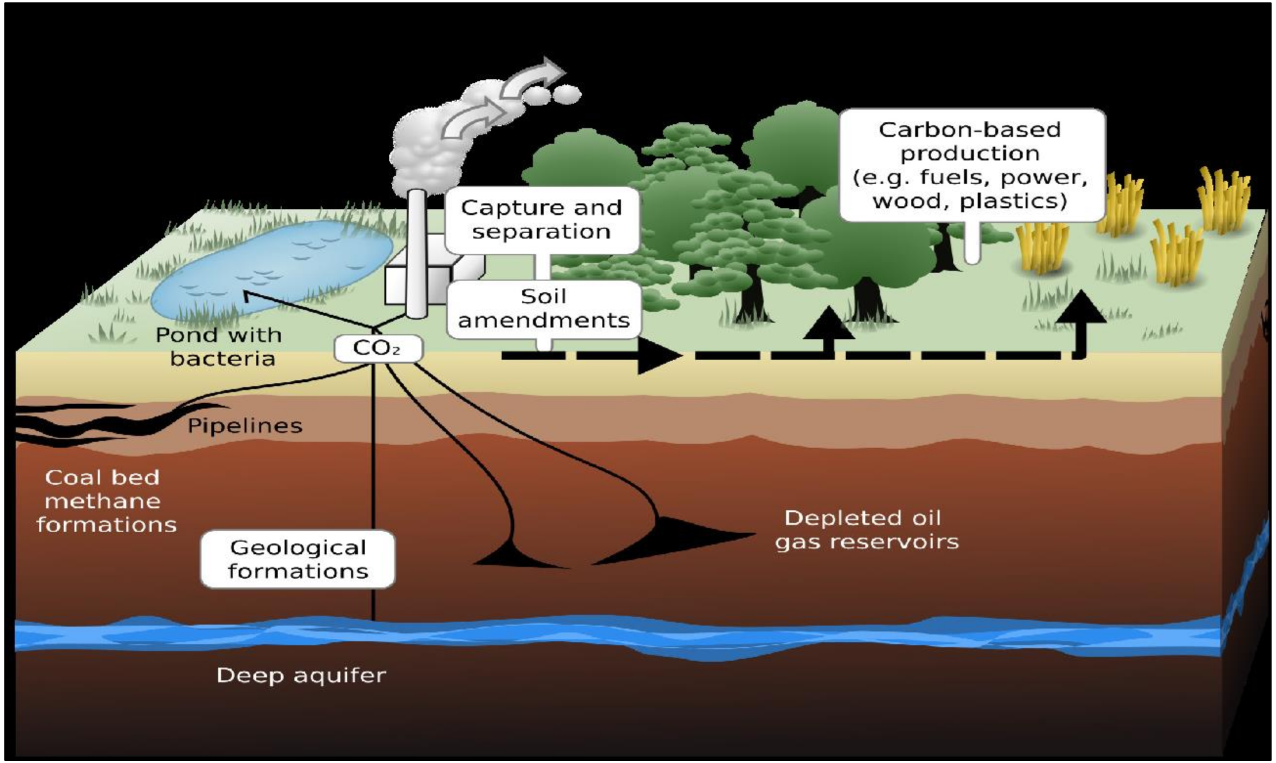
- **Carbon Sink and Carbon Sequestration**
- A **carbon sink** is a natural or artificial reservoir that accumulates and stores some carbon-containing chemical compound indefinitely.
- The process by which carbon sinks remove carbon dioxide (CO2) from the atmosphere is known as **carbon sequestration.**
- Natural carbon sinks include forests, soil and oceans.
- Unlike the other natural sinks, ocean uptake of carbon dioxide results in acidification, threatening species like corals.

Carbon Sink vs Carbon Source

- A **carbon sink** is anything that absorbs more carbon than it releases, whilst a **carbon source** is anything that releases more carbon than it absorbs.
- Forests, soils, oceans and the atmosphere all store carbon, and this carbon moves between them in a continuous cycle (carbon cycle).
- This constant movement of carbon means that forests, soils, atmosphere and oceans act as sources or sinks at different times.
- Recall our discussion on **carbon cycle**

Sustainable Development



- **Forests as carbon Sinks**
- Trees absorb CO_2 during photosynthesis thereby converting atmospheric CO_2 into biomass. When this biomass is buried, the carbon is trapped, forming a carbon sink. When the carbon sink is exposed, the biomass decomposes, adding methane to the atmosphere.
- When biomass is used as fuel (coal, petroleum), it releases CO_2 into the atmosphere (carbon source).
- Forests are carbon dioxide sinks when they increase in density or area and become carbon source when they are degraded (forest degradation).

Oceans as Carbon Sink

- **Blue carbon** is the term for carbon captured by the world's ocean and coastal ecosystems (seagrasses, mangroves, salt marshes, etc.).
- The coastal systems, though much smaller in size than the planet's forests, sequester carbon at a much faster rate, and can continue to do so for millions of years. When these systems are damaged, an enormous amount of carbon is emitted back into the atmosphere.

PYQ: Q1

2017

In the context of mitigating the impending global warming due to anthropogenic emissions of carbon dioxide, which of the following can be the potential sites for carbon sequestration?

1) Abandoned and uneconomic coal seams

2) Depleted oil and gas reservoirs

3) Subterranean deep saline formations

Select the correct answer using the code given below:

a) 1 and 2 only

b) 3 only

c) 1 and 3 only

d) 1, 2 and 3

PYQ: Q1

Answer: d) all

Explanation:

- Above figure shows that abandoned coal seams, depleted oil and gas reservoirs can be used for carbon sequestration.

PYQ: Q2

2014

The scientific view is that the increase in global temperature should not exceed 2°C above pre-industrial level. If the global temperature increases beyond 3°C above the preindustrial level, what can be its possible impact/impacts on the world?

1) Terrestrial biosphere tends toward a net carbon source

2) Widespread coral mortality will occur.

3) All the global wetlands will permanently disappear.

4) Cultivation of cereals will not be possible anywhere in the world.

Select the correct answer using the code given below.

a) 1 only

b) 1 and 2 only

c) 2, 3 and 4 only

d) 1, 2, 3 and 4

PYQ: Q2

Answer: b) 1 and 2 only

• Taiga and temperate forests act as an essential carbon sink. Global warming by 3°C will turn these forests into carbon source (because of the thawing of permafrost and wildfires).

• Corals are overly sensitive to temperature changes. 3°C rise in global temperature will lead to widespread coral mortality.

• It has been estimated that a sea-level rise of approximately 2.3 metres for each degree Celsius of temperature can occur within the next 2,000 years.

PYQ: Q2

- 3°C rise in global temperature will lead to the submergence of many low-lying coastal wetlands like Sundarbans, Chilika Lake, etc., due to the rise in sea levels. Inland wetlands like Keoladeo Ghana National Park, Kolleru Lake, etc., will not be affected.
- Cultivation of cereals in the tropics will take a hit. But in temperate regions, their production increases in the short run.

PYQ: Q3

2021

What is blue carbon?

a) Carbon captured by oceans & coastal ecosystems

b) Carbon sequestered in forest biomass & agricultural soils

c) Carbon contained in petroleum & natural gas.

d) Carbon present in the atmosphere

PYQ: Q3

Answer: A Carbon captured by oceans & coastal ecosystems

Blue carbon is the term for carbon captured by the world's ocean and coastal ecosystems (seagrasses, mangroves, salt marshes, etc.).

PYQ: Q4

2016

'Net metering' is sometimes seen in the news in the context of promoting the

a) production and use of solar energy by the households/consumers

b) use of piped natural gas in the kitchens of households

c) installation of CNG kits in motor-cars

d) installation of water meters in urban households

PYQ: Q4

Answer: A)

Explanation: Net Metering

• Net Metering is a billing mechanism for grid connected Home Rooftop Solar Installation where the electricity generated by the solar panels is fed into the utility grid Household draws electricity from the utility grid

• The household pays only for the difference between the energy units it consumes from the grid and the energy units fed into the grid. This is measured by a bi-directional meter called Net Meter.

PYQ: Q5

2016

Consider the following statements:

1) The International Solar Alliance was launched at the United Nations Climate Change Conference in 2015.

2) The Alliance includes all the member countries of the United Nations.

Which of the statements given above is/are correct?

a) 1 only

b) 2 only

c) Both 1 and 2

d) Neither 1 nor 2

PYQ: Q5

Answer: A) 1 only

- The International Solar Alliance (ISA) is an alliance of more than **122 countries** (data keeps varing) initiated by India, most of them being sunshine countries, which lie either completely or partly between the Tropic of Cancer and the Tropic of Capricorn(now extended to all members of the UN).

The primary objective of the alliance is to work for efficient exploitation of solar energy to reduce dependence on fossil fuels.

This initiative was first proposed by Indian Prime Minister Narendra Modi in a speech in November 2015 at Wembley Stadium, in which he referred to sunshine countries as Suryaputra ("Sons of the Sun").

The alliance is a treaty-based intergovernmental organization. Countries that do not fall within the Tropics can join the alliance and enjoy all benefits as other members, with the exception of voting rights.

PYQ: Q6

2016

On which of the following can you find the BEE Star Label?

a) Ceiling fans

b) Electric geysers

c) Tubular fluorescent lamps

Select the correct answer using the code given below.

a) 1 and 2 only

b) 3 only

c) 2 and 3 only

d) 1, 2 and 3

PYQ: Q6

Answer: d) all

We can find the Bureau of Energy Efficiency Star on Label Ceiling fans, Electric geysers, Tubular fluorescent lamps etc.

- The Bureau initiated the Standards & Labeling programme for equipment and appliances in 2006 to provide the consumer an informed choice about the energy saving and thereby the cost saving potential of the relevant marketed product. The scheme is invoked for 21 equipment/appliances, i.e. Room Air Conditioners, **Tubular Fluorescent Tube Lights,** Frost Free Refrigerators, Distribution Transformers, Induction Motors, Direct Cool Refrigerator, **electric storage type geyser, Ceiling fans,** Color TVs, Agricultural pump sets, LPG stoves, Washing machine, Laptops, ballast, floor standing ACs, office automation products, Diesel Generating sets & Diesel pumpsets.

PYQ: Q7

2014

With reference to technologies for solar power production, consider the following statements:

1. ‘Photovoltaics’ is a technology that generates electricity by direct conversion of light into electricity, while ‘Solar Thermal’ is a technology that utilizes the Sun’s rays to generate heat which is further used in electricity generation process.

2. Photovoltaics generates Alternating Current (AC), while Solar Thermal generates Direct Current (DC).

3. India has manufacturing base for Solar Thermal technology, but not for Photovoltaics.

Which of the statements given above is / are correct?

a) 1 only

b) 2 and 3 only

c) 1, 2 and 3

d) None

PYQ: Q7

Answer: a) 1 only

• Photoelectric effect > When light strikes on a suitable semiconductor material, electrons are dislodged (photons dislodge electrons)

• Photovoltaic > The dislodged electrons, if channelled through a conductor, will create electric current (voltage or potential difference) — electric current is nothing but the movement of electrons from high potential to low potential area (more electrons to fewer electrons region))

• Rotating > AC, Stationary > DC.

Electric generators and wind turbines generate AC while solar panels generate DC.

• Solar thermal > converting light into heat. E.g., solar cooker and solar water heater. Electricity can be generated by using hot water steam to rotate the turbine> AC current.

PYQ: Q8

2016

Consider the following statements:

1) The SDGs were first proposed in 1972 by a global think tank called the 'Club of Rome'.

2) The SDGs have to be achieved by 2030.

Which of the above statements is/are correct?

a) 1 only

b) 2 only

c) Both 1 and 2

d) Neither 1 nor 2

PYQ: Q8

Answer: b) 2 only

- Colombia proposed the idea of the SDGs at a preparation event for Rio+20 held in Indonesia in July 2011
- The Sustainable Development Goals are the blueprint to achieve a better and more sustainable future for all.
- They address the global challenges we face, including those related to poverty, inequality, climate, environmental degradation, prosperity, and peace and justice, the consensus is that , **these SDGs have to be achieved by 2030.**

PYQ: Q9

2016

Which of the following best describes/describe the aim of 'Green India Mission' of the Government of India?

1) Incorporating environmental benefits and costs into the Union and State Budgets thereby implementing the `green accounting'

2) Launching the second green revolution to enhance agricultural output so as to ensure food security to one and all in the future

3) Restoring and enhancing forest cover and responding to climate change by a combination of adaptation and mitigation measures

Select the correct answer using the code given below.

a) 1 only

b) 2 and 3 only

c) 3 only

d) 1, 2 and 3

PYQ: Q9

Answer: c) 3 only

- The National Mission for Green India (GIM) aims at protecting; restoring and enhancing India's diminishing forest cover and responding to climate change by a combination of adaptation and mitigation measures.
- It envisages a holistic view of greening and focuses on multiple ecosystem services, especially, biodiversity, water, biomass, preserving mangroves, wetlands, critical habitats etc. along with carbon sequestration as a co-benefit.
- This mission has adopted an integrated cross-sectoral approach as it will be implemented on both public as well as private lands with a key role of the local communities in planning, decision making, implementation and monitoring.

PYQ: Q10

2011-12

Government of India encourages the cultivation of 'sea buckthorn'. What is the importance of this plant?

1) It helps in controlling soil erosion and desertification.

2) It is a rich source of biodiesel.

3) It has nutritional value and is well-adapted to live in cold areas of high altitudes.

4) Its timber is of great commercial value.

Which of the statements given above is /are correct?

a) 2, 3 and 4 only

b) 1 and 3 only

c) 1, 2, 3 and 4

d) 1 only

PYQ: Q10

Answer: b) 1 and 3 only

- Seabuckthorn has multi-purpose medicinal and nutritional properties. The plant has the ability to fix atmospheric nitrogen.

- It is tolerant to extreme temperatures and has an extensive root system, making it ideal for controlling soil erosion and preventing desertification.

PYQ: Q11

2020

With reference to solar water pumps, consider the following statements:

1. Solar power can be used for running surface pumps and not for submersible pumps,
2. Solar power can be used for running centrifugal pumps and not the ones with piston.

Which of the statements given above is/are correct?

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

PYQ: Q11

Answer D) Neither 1 nor 2

- Solar water pumps are specially designed to utilize DC electric power from photovoltaic modules.
- The pumps must work during low light conditions, when power is reduced, without stalling or overheating.
- Low volume pumps use positive displacement (volumetric) mechanisms which seal water in cavities and force it upward. Lift capacity is maintained even while pumping slowly. These mechanisms include diaphragm, vane and piston pumps. These differ from a conventional centrifugal pump that needs to spin fast to work efficiently.
- Centrifugal pumps are used where higher volumes are required. **So statement 2 is not correct.**
- A surface pump is one that is mounted at ground level. Surface pumps work well when they draw water through suction less than 10 or 20 feet. A submersible pump is one that is lowered into the water. Most deep wells use submersible pumps. And both are compatible with the photovoltaic array (For Solar power). **So statement 1 is not correct.**

PYQ: Q12

2022

Consider the following statements:

1. Gujarat has the largest solar park in India.
2. Kerala has a fully solar powered International Airport.
3. Goa has the largest floating solar photovoltaic project in India.

Which of the statements given above is/are correct?

(a) 1 and 2
(b) 2 only
(c) 1 and 3
(d) 3 only

PYQ: Q12

Answer: B

- **Statement 1 is incorrect:** Rajasthan tops the list of solar park installations in the country followed by Karnataka and then Andhra Pradesh. **Bhadla Solar Park in Rajasthan, with a capacity of 2245 MW, is the world's largest solar park.**

- **Statement 2 is correct: Cochin International Airport, India's first airport built under a public-private-partnership (PPP) model, becomes the first airport in the world that operates completely on solar power.** This plant is the first Megawatt scale installation of a Solar PV system in the State of Kerala

- **Statement 3 is incorrect: Andhra Pradesh was the home to India's largest floating solar power plant.** State-run NTPC started operations at India's largest floating solar PV project at its Simhadri thermal station in Visakhapatnam. **(NTPC has announced largest floating solar power plant in Ramagundam, Telangana recently)**

PYQ: Q13

2017

Consider the following statements:

The nationwide 'Soil Health Card Scheme' aims at

1. expanding the cultivable area under irrigation.
2. enabling the banks to assess the quantum of loans to be granted to farmers on the basis of soil quality.
3. checking the overuse of fertilizers in farmlands.

Which of the above statements is/are correct?

a) 1 and 2 only
b) 3 only
c) 2 and 3 only
d) 1, 2 and 3

PYQ: Q13

Ans.14) B

- Soil Health Card (SHC) is a Government of India's scheme promoted by the Department of Agriculture & Cooperation under the Ministry of Agriculture and Farmers' Welfare.
- Launched in February 2015, the scheme is tailor-made to issue a 'Soil card' to farmers which will carry crop-wise recommendations of nutrients and fertilizers required for the individual farms. This is aimed to help farmers to improve productivity through the judicious use of inputs.